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Vasoactive Molecules and the Kidney

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CHAPTER OUTLINE

KEY POINTS

- Beyond the classical endocrine renin–angiotensin system (RAS) that pivots around the actions of systemically circulating angiotensin II, an additional, quasi-independent local RAS operates within the kidney in paracrine, autocrine, and possibly even intracrine modes.
- Although much of our understanding of the RAS relates to the actions of angiotensin II, a range of other angiotensin-related proteins, enzymes, and receptors also have important biological actions in both kidney physiology and disease.
- Therapies that block the RAS have been employed to slow renal decline in chronic kidney disease for decades and agents that either prevent (e.g., endothelin receptor antagonists) or augment (e.g., neprilysin inhibitors in combination with angiotensin receptor blockade) the actions of other vasoactive molecules are under investigation.
- Endothelin type A receptor blockade is undergoing phase III evaluation for the treatment of diabetic nephropathy, although earlier clinical trials were hampered by dose-dependent fluid retention.
- Inhibition of neprilysin prevents the degradation of the natriuretic peptides atrial natriuretic peptide and brain natriuretic peptide, but its effects are limited by an unopposed RAS. Concurrent angiotensin receptor blockade and neprilysin inhibition improves outcomes in patients with heart failure, although its effects on hard renal outcomes are currently unknown.
- Concurrent angiotensin-converting enzyme inhibition and neprilysin inhibition are associated with an increased risk of angioedema.
- The development of therapies that affect other vasoactive molecules (e.g., the kallikrein–kinin system, urotensin II, guanylin, uroguanylin, and adrenomedullin) has been limited.

Vasoactive peptides, arising from both the systemic circulation and from local tissue-based generation, play important roles in kidney physiology, not only in the regulation of renal blood flow (RBF) but also in electrolyte exchange, acid–base balance, and diuresis. More recent interest has focused on the role of these peptide systems in kidney development and in the pathogenesis of organ injury.

RENIN–ANGIOTENSIN–ALDOSTERONE SYSTEM

In their now seminal 1898 report, *Niere und Kreislauf*, Robert Tigerstedt and Per Bergman, while working at the Karolinska Institute in Sweden, described the prolonged vasopressor effects of crude kidney extracts.[1] Although recognizing the impurity of the extract, Tigerstedt named the unidentified active substance, "renin," based on its organ of origin. More than 110 years later, our understanding of the renin–angiotensin system (RAS) continues to evolve with recent insights into its pivotal role in pathophysiological as well as physiological processes. Underlying this effort to fully understand the RAS is not only a desire for knowledge but also a profound appreciation of the therapeutic importance of its blockade that emanates from the renoprotective effects of angiotensin-converting enzyme inhibition first described by Anderson, Meyer, Rennke, and Brenner in 1985 in a rodent model of progressive kidney disease.[2]

Clinical Relevance

The renin–angiotensin system (RAS) plays a fundamental role in blood pressure, plasma volume, electrolyte, and acid–base homeostasis. Beyond its function in kidney physiology, however, angiotensin II, the primary effector molecule of the RAS, raises intraglomerular pressure, induces proteinuria, and stimulates the production of extracellular matrix that leads to glomerulosclerosis and interstitial fibrosis. Accordingly, blockade of angiotensin II synthesis by angiotensin-converting enzyme inhibition or antagonizing its action at the angiotensin I receptor with an angiotensin receptor blocker is at the cornerstone of strategies that attenuate progressive kidney function decline in most forms of chronic kidney disease.

CLASSICAL RENIN–ANGIOTENSIN–ALDOSTERONE SYSTEM

The classical view of the RAS focuses on the endocrine aspects of this peptidergic system. Angiotensinogen synthesized by the liver enters the circulation where it is cleaved to form angiotensin I by renin, a peptidase that is secreted from the juxtaglomerular apparatus (JGA) of the kidney. The terminal two amino acids of angiotensin I are then removed to form angiotensin II, as it traverses through the circulation, exposed to angiotensin-converting enzyme (ACE), a peptidase robustly expressed on endothelial cells of the pulmonary vasculature. Angiotensin II, the principal effector molecule of the RAS, then binds to its type 1 receptor (AT_1R), resulting in vasoconstriction, sodium retention, thirst, and aldosterone secretion. This traditional view of the RAS is still valid but has been considerably augmented in recent years, not only by the discovery of new enzymes, peptides, and receptors, but also by an appreciation that the RAS has an independently functioning local tissue–based component that acts through paracrine, autocrine, and possibly intracrine mechanisms (Fig. 11.1).

ANGIOTENSINOGEN

Angiotensinogen is primarily, although by no means exclusively, synthesized in the liver, particularly the pericentral zone of the hepatic lobules.[3] In humans, it is coded by a single gene, composed of five exons and four introns, that spans about 13 kb of genomic sequence on chromosome 1 (1q42-q43). It is translated to a 453 amino acid globular glycoprotein with a molecular weight between 45 and 65 kDa, depending on the extent of its glycosylation, that then undergoes posttranslational cleavage of a 24– or 33–amino acid signal peptide,[4] giving rise to the mature circulating form of angiotensinogen.[5]

Structurally, angiotensinogen bears substantial homology to the serpin superfamily of protease inhibitors and like many members of its family behaves as an acute phase reactant in the inflammatory setting,[6] reflecting the presence of an acute-phase response element that binds the transcription factor, NF-κB (nuclear factor kappa-light-chain-enhancer of activated B cells).[7]

RENIN

Like angiotensinogen, the gene encoding renin is also located on the long arm of chromosome 1 (1q32) and contains 10 exons and 9 introns, similar to other aspartyl proteases.[8] Unlike humans and rats that have only a single renin gene, the mouse has two genes, *Ren1* and *Ren2*, expressed primarily in the submandibular gland and kidney, respectively.

Following its synthesis as a 406–amino acid preprohormone, the 23–amino acid leader sequence of preprorenin is cleaved in the rough endoplasmic reticulum, giving rise to prorenin (also called inactive renin and "big" renin), which may be then rapidly secreted directly from the Golgi apparatus or from protogranules.[4] Alternatively, and virtually exclusively in the JGA, prorenin may be packaged into mature, dense granules that instead of being immediately secreted, undergo further processing to the active enzyme, renin (active renin). Contrasting with the more constitutive secretion of prorenin, the release of renin-containing granules is tightly regulated.[8]

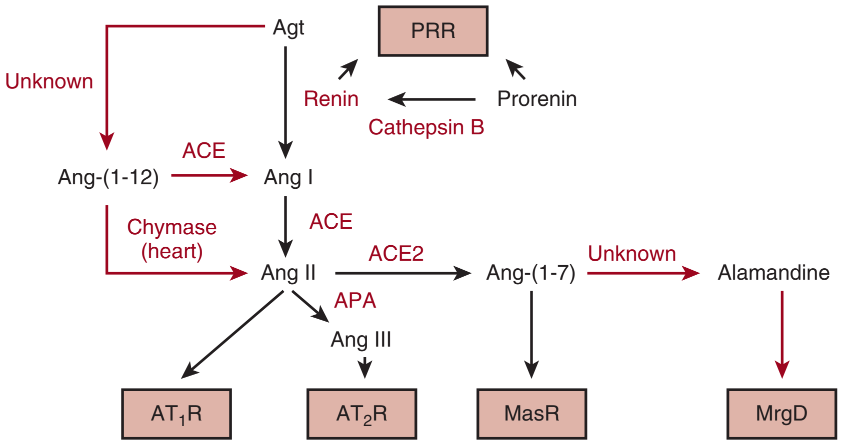

Fig. 11.1 Schematic depiction of the renin–angiotensin system components and selected actions. The enzymes of the system are shown in *red.* Newly described enzymatic pathways are shown as *red arrows.* Receptors are shown in the *boxes*. *ACE,* Angiotensin-converting enzyme; *Agt,* angiotensinogen; *Ang,* angiotensin; *APA,* aminopeptidase A; *AT_1R,* angiotensin type-1 receptor; *AT_2R,* angiotensin type-2 receptor; *MasR, Mas* receptor; *MrgD, Mas*-related G-protein–coupled receptor; *PRR,* (pro)renin receptor. (Modified from Carey RM. Newly discovered components and actions of the renin-angiotensin system. *Hypertension.* 2013;62: 818–822.)

Mature, active renin is a variably glycosylated 340–amino acid 37–40-kDa aspartyl protease that is active at neutral pH, and in contrast to the more promiscuous activities of most other proteases in this class, has only a single known substrate, cleaving the decapeptide angiotensin I from the amino terminal of angiotensinogen. Although the kidney produces both renin and prorenin, a range of extrarenal tissues including the adrenals, gonads, and placenta produce prorenin and contribute to its presence in plasma. However, as evidenced by the near total absence of active renin in anephric patients, the kidney, and the JGA in particular, appears to be the only source of circulating renin in humans.

Factors that chronically stimulate renin secretion, such as a low sodium diet and ACE inhibition, lead to an increase in the number of renin secreting cells rather than an increase in cell size or the number of granules that each JGA cell contains. This expansion of the renin secreting mass occurs proximally by metaplastic transformation of smooth muscle cells within the walls of the afferent arteriole. Although sometimes mentioned, ectopic renin expression within the extraglomerular mesangium appears to be an uncommon event.[9]

PRORENIN ACTIVATION

Prorenin is maintained as an inactive zymogen through the occupation of its catalytic cleft by its prosegment. Removing this prosegment by either proteolytic or nonproteolytic means yields active renin, a term that denotes its enzymatic activity rather than its amino acid sequence (Fig. 11.2).

Within the dense core secretory granules of the JGA, acidification by vacuolar adenosine triphosphatases (ATPases) provides the optimal pH for the prosegment cleaving enzymes (proconvertase 1 and cathepsin B), and may also assist the pH-dependent, nonenzymatic activation of prorenin as well.[9–11] Although various peptidases such as trypsin, plasmin, and kallikrein can also cleave the prosegment of prorenin in vitro, these do not appear to contribute to the generation of renin in the in vivo setting. Although traditionally viewed as occurring only in the JGA, recent cell culture–based studies suggest that proteolytic activation of renin can also occur in cardiac and vascular smooth muscle cells by as yet unidentified serine proteases.[12–14] The significance of these findings in the intact organism, however, remains to be established.

In addition to proteolytic cleavage of its prosegment, prorenin can also be reversibly activated nonenzymatically by a conformational change such that the prosegment no longer occupies the enzymatic cleft. Under usual circumstances, less than 2% of prorenin is in this open active conformation. This process can, however, be induced by acid (pH 4.0)[15,16] and to a lesser extent by cold.[17] More recently, the putative (pro)renin receptor (PRR; discussed later) has also been shown to nonproteolytically activate prorenin.[18]

REGULATION OF RENIN SECRETION

Mechanical, neurological, and chemical factors regulate the activity of the RAS by modulating renin secretion.

Renal Baroreceptor

The existence of a renal baroreceptor mechanism was first conceptualized by Skinner and colleagues to explain how renin secretion increases when afferent arteriolar perfusion pressure falls.[19] Studies in conscious dogs show that changes in renal perfusion pressure have only a small effect on renin secretion until a threshold of about 90 mm Hg is reached, below which renin secretion abruptly increases, doubling with every 2 to 3 mm Hg fall in pressure.[20] Accordingly, reduction in pressure below this level profoundly stimulates renin secretion, thereby acutely activating the RAS and resulting in a range of angiotensin II–dependent phenomena that collectively serve to restore systemic pressure. Despite the importance of the baroreceptor function, several decades of research have not identified precisely how the pressure signal is transduced into renin release, though postulated mediators include stretch-activated calcium channels, endothelins (ETs), and prostaglandins.

Neural Control

The JGA is endowed with a rich network of noradrenergic nerve endings and their β1 receptors. Stimulation of the

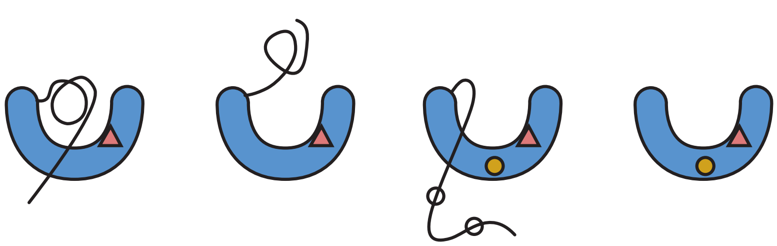

Fig. 11.2 The conformational changes and the expression of immunoreactive epitopes associated with the activation of prorenin are depicted. The main body of the molecule *(blue)*, the substrate-binding cleft, and the prosegment *(black line)* are shown. The closed triangle represents the epitope of the main body expressed by PR_c (prorenin in the inactive closed conformation), PR_{oi} (prorenin in the inactive intermediary open conformation), PR_o (prorenin in the active open conformation), and renin. The closed circle *(yellow)* represents the epitope of the main body, expressed by PR_o and renin, but not by PR_c and PR_{oi}. The *open circles* represent epitopes of the prosegment expressed by PR_o but not by PR_c and PR_{oi}. (Modified from Schalekamp MA, Derkx FH, Deinum J, et al. Newly developed renin and prorenin assays and the clinical evaluation of renin inhibitors. *J Hypertension.* 2008;26:928–937.)

renal sympathetic nerve activity leads to renin secretion that is independent of changes in RBF, glomerular filtration rate (GFR), or Na^+ reabsorption. Moreover, this effect can be blocked surgically (denervation) and pharmacologically, by the administration of β-adrenoreceptor blockers.[20] The role of cholinergic, dopaminergic, and adrenergic activation is controversial, though these agents have also been shown to modulate renin release under certain circumstances.

Tubule Control

Chronic diminution in luminal NaCl delivery to the macula densa is a potent stimulus for renin secretion, reflecting a coordinate interaction between a range of mediators including adenosine, nitric oxide (NO), and prostaglandins that not only affect renin release but also its transcription.[21] This mechanism is thought to account for the chronically high plasma renin activity (PRA) in subjects ingesting a low-salt diet.[22]

Metabolic Control

The tricarboxylic acid (TCA) cycle provides a final common pathway by which carbohydrates, fatty acids, and amino acids converge in the process of adenosine triphosphate (ATP) generation by aerobic electron transfer. Although the TCA cycle operates within mitochondria, its intermediates can be detected within the extracellular space, increasing in abundance when local energy supply and demand are mismatched or when cells are exposed to hypoxia, toxins, or injury.[23] Succinate, for instance, has been shown to stimulate renin release and its intravenous administration leads to hypertension, though the mechanisms underlying this effect have only recently been unraveled. In 2004, He and colleagues reported that alpha keto-glutarate and succinate are ligands for the previously orphaned G protein–coupled receptors (GPCRs), GPR99 and GPR91, respectively, and that succinate-induced hypertension is abolished in GPR91-deficient mice.[24] Indeed, in follow-up studies from this group, GPR91 was localized to the apical plasma membrane of macula densa cells where succinate stimulation was shown to activate p38 and Erk 1/2 mitogen-activated protein (MAP) kinases (MAPKs), inducing cyclooxygenase-2 (COX-2)-dependent synthesis of prostaglandin E2, a well-established paracrine mediator of renin release.[25] Moreover, the ability of tubule succinate to induce JG renin secretion suggests that this phenomenon is likely an important determinant of JGA function in both physiological and pathophysiological settings. In diabetic rats, for instance, elevated succinate has been detected in both plasma and urine.[25]

Vitamin D Receptor

The vitamin D receptor (VDR) is a negative regulator of the RAS such that VDR-null mice display a marked increase in renin expression and angiotensin II production in conjunction with hypertension and cardiac hypertrophy.[26] Importantly, these effects occurred independently of calcium and parathyroid hormone, both of which have been reported to also modulate renin expression. The molecular basis for the interaction between vitamin D and renin expression has also been, at least partly, unraveled in a series of studies by Yuan et al.[27] Under usual circumstances, activation of the cyclic adenosine monophosphate (cAMP)–protein kinase A pathway by the sympathetic nervous system or macula densa leads to phosphorylation of a cAMP response element (CRE) binding protein (CREB) and recruitment of CREB-binding protein/p300 to the CRE in the promoter region of the renin gene. VDR-bound 1,25 $(OH)_2D_3$, however, blocks binding of CREB to the CRE DNA *cis*-element, leading to a reduction in renin gene transcription (Fig. 11.3).

From a clinical perspective, this interaction between vitamin D and renin may explain the well-documented inverse relationship between plasma vitamin D_3, blood pressure, and PRA. While several studies examining the effects of vitamin D supplementation as a renoprotective or antihypertensive measure have been undertaken, their findings have been mixed and long-term, randomized controlled trials with clinically meaningful endpoints are awaited.

Other Local Factors

In addition to the factors discussed earlier, a large range of locally produced biologically active molecules have also been shown to alter renin secretion. These include peptides [ANP, kinins, vasoactive intestinal polypeptide, ET, calcitonin gene–related peptide (CGRP)], amines (dopamine and histamine), and arachidonic acid derivatives.[20]

PLASMA PRORENIN AND RENIN

Under usual circumstances, the plasma concentration of prorenin is approximately 10 times greater than renin. In some patients with diabetes, however, plasma prorenin is disproportionately increased, where it predicts the development of diabetic nephropathy (including microalbuminuria) and retinopathy.[28,29]

In addition to its role in the research setting, measurement of plasma renin is an important clinical assay, providing important information, for example, when evaluating patients with possible hyperaldosteronism, assessing volume status, and in predicting the response to, or monitoring drug

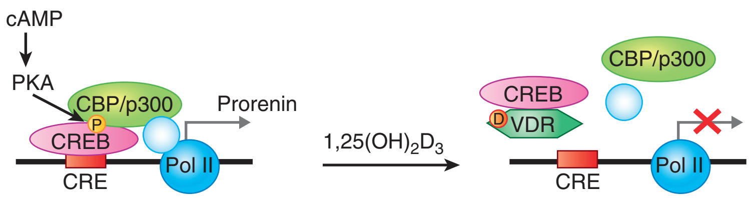

Fig. 11.3 Model of $1,25(OH)_2D_3$-induced transrepression of renin gene expression. The cAMP–PKA pathway activates CREB by phosphorylation, leading to recruitment of CBP/p300. In the presence of $1,25(OH)_2D_3$, liganded VDR interacts with CREB and blocks its binding to CRE, leading to reduction of renin gene transcription. *cAMP,* Cyclic AMP; *CBP,* CREB-binding protein; *CRE,* cAMP response element; *CREB,* CRE-binding protein; *D,* $1,25(OH)_2D_3$; *P,* phosphorylation; *PKA,* protein kinase A; *Pol II,* RNA polymerase II; *VDR,* vitamin D receptor. (Modified from Yuan W, Pan W, Kong J, et al. 1,25-Dihydroxyvitamin D_3 suppresses renin gene transcription by blocking the activity of the cyclic AMP response element in the renin gene promoter. *J Biol Chem.* 2007;282:29821–29830.)

adherence to, an ACE inhibitor or angiotensin receptor blocker (ARB). In broad terms, plasma renin is determined by either activity or immunological assay methods.[30] The most commonly used method involves the measurement of PRA. With this method, the rate at which angiotensin I is produced from plasma angiotensinogen is assayed. To prevent angiotensin I's degradation or its conversion to angiotensin II, inhibitors of angiotensinase and ACE are added to the assay. Accordingly, PRA is not only dependent on renin and endogenous angiotensinogen concentrations but will also overestimate the extent of inhibition by renin inhibitors due to the displacement of protein-bound drug by the peptidase inhibitors. The latter scenario may be diminished by using an antibody capture method in which antiangiotensin I antibody, instead of peptidase inhibitors, is used to protect angiotensin I from further catabolism.[31]

The nomenclature of renin assays can be quite confusing in that plasma renin concentration (PRC) may be measured by both activity and immunological assays. With the activity method (PRCa), exogenous angiotensinogen is added to the assay, thereby avoiding the influence that endogenous levels of the substrate might have. However, PRCa may also be affected by the presence of renin inhibitors, though like PRA may take advantage of antibody capture methodology.[30] In the immunological assay for renin (PRCi), the concentrations of renin and prorenin in its active, open conformation are assessed, so that like PRA and PRCa, the PRCi assay is also time and temperature dependent because lower temperatures will increase the proportion of prorenin in its active conformation. Moreover, renin inhibitors, by binding to the active site of prorenin in its open conformation, prevent the refolding of the prosegment and may therefore lead to an overestimation of PRCi.[30,31]

ANGIOTENSIN-CONVERTING ENZYME

ACE is a zinc-containing dipeptidyl carboxypeptidase that cleaves the terminal histidyl-leucine from angiotensin I to form the octapeptide angiotensin II. In contrast to the single-substrate specificity of renin, ACE is not specific, cleaving the two terminal acids from peptides with the C′-terminal sequence R_1-R_2-R_3-OH, where R_1 is the protected (noncleaved) amino acid, R_2 is any nonproline L-amino acid, and R_3 is any nondicarboxylic (cysteine, ornithine, lysine, arginine) L-amino acid with a free carboxyl terminal.[4] Importantly, therefore, ACE also catalyzes the inactivation of bradykinin. Although encoded by a single ACE gene, two distinct tissue-specific messenger RNAs (mRNAs) are transcribed, each with different initiation and alternative splice sites.[32] The somatic form, present in almost all tissues, is a 1306–amino acid, 140–160-kDa glycoprotein with two active sites, whereas the 90- to 100- kDa testicular or germinal form is found exclusively in postmeiotic male germ cells, and contains a single active site and appears to be involved in spermatogenesis.[4,33,34] The somatic form of ACE is widely distributed with activity present not only in tissues but also in most biological fluids. In the human kidney, ACE is present to the greatest extent within proximal and distal tubules, however, both its magnitude of expression and its site-specific distribution may be altered by disease.[35]

ANGIOTENSIN TYPE 1 RECEPTOR

The AT_1R mediates most of the known physiological effects of angiotensin II. The gene for this widely distributed 359–amino acid, 40-kDa, seven-transmembrane GPCR is located on chromosome 3 in humans.[36]

Within the kidney, AT_1Rs are widely expressed. In the glomerulus, they are found in both afferent and efferent arterioles as well as in the mesangium, endothelium, and on podocytes.[37] Consistent with angiotensin II's role in Na^+ reabsorption, AT_1Rs are highly abundant on the brush borders of proximal tubule epithelial cells.[38] Prominent expression has also been found in renal medullary interstitial cells, located between the renal tubules and vasa recta, where angiotensin II is purported to have a potential role in the regulation of medullary blood flow.[39]

Angiotensin II binding to AT_1R initiates cell signaling by several different pathways that have been mostly studied in vascular smooth muscle cells.[40] These include G protein–mediated pathways, and the activation of tyrosine kinases, NADH/NADPH oxidases, and serine/threonine kinases.[36]

G Protein–Mediated Signaling

In the classical G protein–mediated pathway, AT_1R ligand binding leads to activation of phospholipases C, D, and A_2. Phospholipase C rapidly hydrolyses phosphatidylinositol bisphosphate to inositol trisphosphate and diacylglycerol (DAG), initiating calcium release from intracellular stores and protein kinase C (PKC) activation, respectively. Phospholipase D similarly generates DAG and activates PKC, whereas phospholipase A2 (PLA_2) leads to the formation of various vasoactive and proinflammatory arachidonic acid derivatives.

Reactive Oxygen Species

Although previously regarded as toxic waste products, emerging evidence indicates that reactive oxygen species may also act as second messengers, not only activating other cell signaling cascades such as p38 MAPK but also a number of transcription pathways implicated in the pathogeneses of inflammatory and degenerative disease.[36] Although the mechanisms by which the AT_1R stimulates NADH/NADPH are not well understood, angiotensin II binding to this receptor results in the generation of both superoxide and hydrogen peroxide.

Tyrosine Kinases

Angiotensin II binding to the AT_1R "transactivates" a number of nonreceptor tyrosine kinases [Src, Pyk2, Focal adhesion kinase (FAK), and Janus kinase (JAK)] as well as the growth factor receptor tyrosine kinases for epidermal growth factor (EGF)[41,42] and platelet-derived growth factor (PDGF).[43,44] By binding to the AT_1R, angiotensin II initiates the translocation of tumor necrosis factor-alpha (TNF-α)–converting enzyme (TACE, ADAM17) to the cell surface. TACE then cleaves TNF-α from its membrane-associated precursor (pro-TNF-α), allowing it to bind to the EGF receptor (EGFR) on the cell surface. This ligand–receptor interaction then induces EGFR autophosphorylation and activates its downstream signaling pathways that include Akt, Erk-1/2, and mammalian target of rapamycin (Fig. 11.4). The in vivo relevance of this transactivation pathway has been recently confirmed. Using mice that express a dominant negative form of EGFR, Lautrette and colleagues showed that despite similar blood pressures, mutant mice infused with angiotensin II had less proteinuria and renal fibrosis than did their wild-type

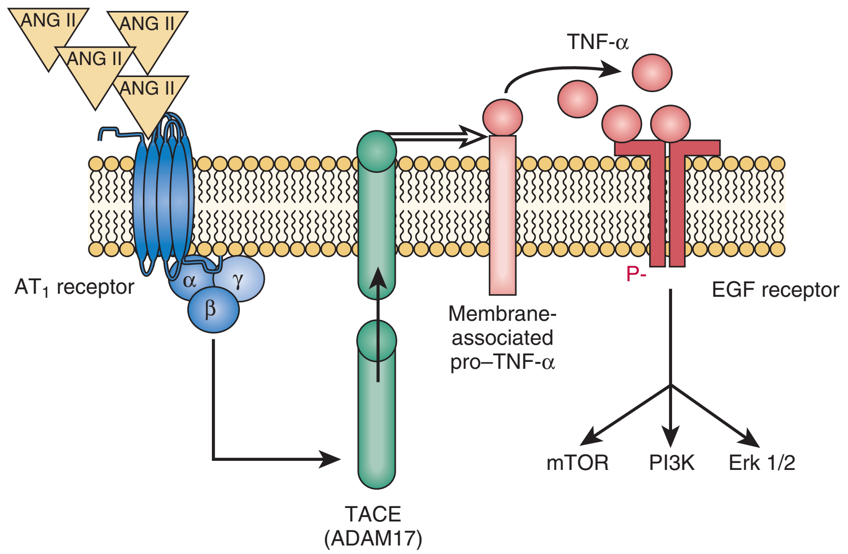

Fig. 11.4 Angiotensin II *(ANG II)* binds to its angiotensin II type 1 *(AT$_1$)* receptor, a G protein–coupled receptor lacking intrinsic tyrosine kinase activity. Through as-yet-undescribed mechanisms, this interaction leads to the translocation of the metalloprotease tumor necrosis factor-α *(TNF-α)*–converting enzyme *(TACE)* from the cytosol to the cell surface, where it cleaves TNF-α from its membrane-associated promolecule, allowing it to bind and activate the epidermal growth factor (EGF) receptor. *Erk 1/2,* Extracellular signal–regulated kinases 1 and 2; *mTOR,* mammalian target of rapamycin; *P13K,* phosphatidylinositol-3-kinase. (Modified from Wolf G. "As time goes by": angiotensin II–mediated transactivation of the EGF receptor comes of age. *Nephrol Dial Transplant.* 2005;20:2050–2053.)

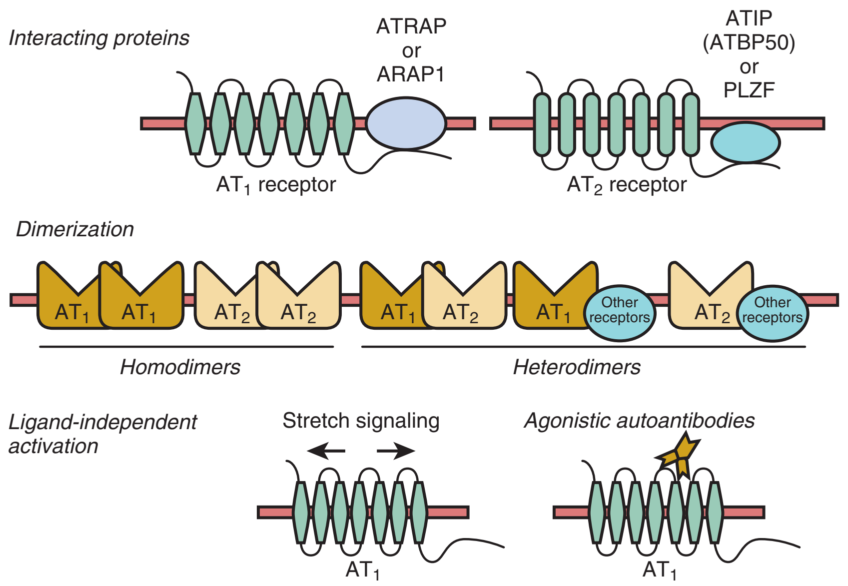

Fig. 11.5 Regulation of angiotensin receptors. *ARAP1,* AT$_1$ receptor–associated protein 1; *ATBP50,* AT$_2$ receptor–binding protein of 50 kDa; *ATIP,* angiotensin type 2 receptor–interacting protein; *AT$_1$ receptor,* angiotensin type 1 receptor; *AT$_2$ receptor,* angiotensin type 2 receptor; *ATRAP,* AT$_1$ receptor–associated protein; *PLZF,* promyelocytic leukemia zinc finger. (Modified from Mogi M, Iwai M, Horiuchi M. New insights into the regulation of angiotensin receptors. *Curr Opin Nephrol Hypertens.* 2009;18:138–143.)

counterparts.[45] Consistent with these findings and the pivotal role of the RAS in diabetic nephropathy, studies using a specific EGFR tyrosine kinase inhibitor (i.e., PKI 166) have also shown a reduction in early structural injury in a rat model of diabetic nephropathy.[46]

Like EGFR, the transactivation of the PDGF receptor (PDGFR) by AT$_1$R is also complex, involving the adaptor protein Shc.[43,44] In addition to studies that have explored the angiotensin–PDGFR interaction in cell culture or organ baths,[47] a more recent report has shown that despite continued hypertension, inhibition of the PDGFR kinase in vivo can also dramatically attenuate angiotensin II–induced vascular remodeling.[48]

Angiotensin Type 1 Receptor Internalization

In addition to the conventional ligand–receptor mediated pathways, a range of other signaling mechanisms that involve the AT$_1$R have also been described. These include the discovery of receptor interacting proteins, heterologous receptor dimerization, and ligand-independent activation (Fig. 11.5).[49]

These new insights, although adding greater complexity to our understanding of the RAS, also provide the potential for new therapeutic targets in disease prevention and management.

Following ligand binding and the initiation of signal transduction, AT_1Rs are rapidly internalized, followed by either lysosomal degradation or recycling back to the plasma membrane. Several mechanisms account for AT_1R internalization, including interaction with caveolae, phosphorylation of its carboxyl terminal by G-protein receptor kinases,[36] and association with the newly described AT_1R interacting proteins.[49] To date, two such interacting proteins, *AT*$_1$ *r*eceptor–*a*ssociated *p*rotein (ATRAP)[50] and *AT*$_1$ *r*eceptor–*a*ssociated *p*rotein-1 (ARAP1),[51] have been described. ATRAP interacts with the C terminal of AT_1R, downregulating cell surface AT_1R expression and attenuating angiotensin II–mediated effects.[49] ARAP1, though somewhat similar to ATRAP, promotes AT_1R recycling to the plasma membrane such that its kidney-specific overexpression induces hypertension and renal hypertrophy.[51]

Angiotensin Type 1 Receptor Dimerization

In addition to their ability to induce cell signaling in their monomeric state, GPCRs like AT_1R may also associate to form both homodimers and heterodimers.[52] Beyond its constitutive homodimerization,[53] AT_1R may dimerize with angiotensin type 2 receptor (AT_2R) and also form hetero-oligomers with receptors for bradykinin (B2), epinephrine (β2), dopamine (D1,3,5), ET (B), Mas, and EGF that modulate their function.[54–57]

Ligand-Independent Angiotensin Type 1 Receptor Activation

Without involvement of angiotensin II, cell stretch induces a conformational switch that initiates AT_1R's intracellular signaling pathways.[58,59] As might be expected from an understanding of this mechanism, an AT_1R blocker, acting as an inverse agonist, will abrogate these effects, as described in both cardiac[59] and mesangial cells.[60] A similar means of ligand-independent activation has also been shown to result from the binding of agonist antibodies to AT_1R in some women with preeclampsia[61] and in certain cases of renal allograft rejection.[62]

PHYSIOLOGIC EFFECTS OF ANGIOTENSIN II IN THE KIDNEY

The traditional actions of angiotensin II relate primarily to its effects on vascular tone and fluid balance that are mediated by its actions on the vasculature, heart, kidney, brain, and adrenal glands by the AT_1R. In vascular smooth muscle, stimulation of AT_1Rs by angiotensin II induces cell contraction and consequent vasoconstriction. In the adrenal cortex, this ligand– receptor interaction stimulates aldosterone release, thereby promoting sodium reabsorption in the distal nephron. Moreover, angiotensin II will directly enhance sodium retention by the proximal tubule and in the brain it will stimulate thirst and salt craving. Additional effects include sympathoadrenal stimulation and the augmentation of cardiac contractility. Together, these effects serve to maintain extracellular fluid volume and systemic blood pressure. Given the central role that the kidney has in the regulation of these key aspects of mammalian homeostasis, it is not surprising that angiotensin II should have profound effects on renal physiology.

HEMODYNAMIC ACTIONS

The effects of exogenously administered angiotensin II are dose dependent. At low doses, angiotensin II infusion increases renal vascular resistance (RVR) and lowers RBF without affecting GFR so that the filtration fraction is increased. At higher doses of angiotensin II, RVR is further increased, leading to an augmented reduction in RBF and fall in GFR.[63] However, because GFR is reduced to a lesser extent than renal plasma flow, the filtration fraction remains elevated. Such studies are consistent with the view that limited stimulation of the RAS would mostly serve to enhance tubule sodium, as is seen, for instance, in societies unaccustomed to contemporary diets.[22] Greater activation of the RAS, by contrast, as might be found in the setting of severe volume depletion, would result in angiotensin II–dependent reduction in RBF that would aid in sustaining systemic blood pressure while further stimulating sodium reabsorption.

Kidney micropuncture has been used extensively to explore the intrarenal sites of angiotensin II's effects on vascular resistance. These studies demonstrate that although angiotensin II increases both afferent and efferent arteriolar resistance, intraglomerular capillary pressure (P_{GC}) is consistently elevated[64] and the ultrafiltration coefficient (K_f) is reduced.[63] Moreover, as predicted by mathematical modeling, the glomerular hypertension induced by angiotensin II does not lead to acute proteinuria, because the structural barriers to macromolecular passage remain intact.[63] Chronic angiotensin II infusion with sustained intraglomerular hypertension, by contrast, leads to glomerular capillary damage and substantial proteinuria.

TUBULE TRANSPORT

Sodium

Consistent with its importance in the regulation of volume status, angiotensin II has profound effects on renal Na^+ handling. The proximal tubule is responsible for the reabsorption of approximately two-thirds of the sodium from the glomerular filtrate and binding sites for angiotensin II are particularly abundant in the proximal tubule with immunohistochemical localization of the AT_1R to both apical and basolateral surfaces.[65] At picomolar concentrations, angiotensin II stimulates the luminal Na^+/H^+ exchanger, the basolateral $Na^+/HCO3^-$ cotransporter, and the $Na^+–K^+$ ATPase. However, at concentrations greater than 10^{-9} M, angiotensin II inhibits the very same transporters. The mechanisms underlying this dose-dependent effect of angiotensin II on Na^+ transport, that seem to also occur in the loop of Henle,[65] are incompletely understood. In the distal tubule, the effects of angiotensin II on Na^+ transport are site dependent. In the early distal tubule, for instance, angiotensin II stimulates apical Na^+/H^+ exchange while in the late distal tubule it stimulates the amiloride-sensitive sodium channel.[65]

Acid–Base Regulation

The kidney has a key role in the maintenance of physiological pH by regulating the secretion/reabsorption of acids and bases. As for Na^+, angiotensin II also has substantial effects on acid–base transport in the proximal tubule, distal tubule, and collecting duct. Recent interest has focused, in particular, on its actions in the collecting duct. In the collecting duct, angiotensin II not only stimulates Na^+/H^+ exchangers and $Na^+/HCO3^-$ cotransporters but has also been shown to stimulate the vacuolar H^+-ATPase in intercalated A-cells via its AT_1R receptor.[66] Moreover, elegant and detailed electron microscopic studies have helped to unravel the mechanisms by which angiotensin II exerts its effects at this site, revealing translocation of the H^+-ATPase from the cytoplasm to the apical surface in response to ligand stimulation.[67]

EXPANDED RENIN–ANGIOTENSIN–ALDOSTERONE SYSTEM: ENZYMES, ANGIOTENSIN PEPTIDES, AND RECEPTORS

ANGIOTENSIN TYPE 2 RECEPTOR

In humans, the AT_2R is a 363–amino acid protein that maps to the X chromosome and is highly homologous to its rat and mouse counterparts.[68] Like AT_1R, AT_2R is also a seven-transmembrane GPCR, though it shares only 34% homology.

Despite substantial research, the actions of AT_2R are still not well understood and remain somewhat controversial.[69] In general, however, the actions of AT_2R stimulation oppose those of AT_1R. For instance, whereas AT_1R vasoconstricts and promotes Na^+ retention, AT_2R stimulation leads to vasodilation[70] and natriuresis,[71] consistent with its abundance on the epithelium of the proximal tubule.[72] The vasodilatory effects of AT_2R stimulation are mediated by increasing NO synthesis and cyclic guanosine monophosphate (cGMP) by bradykinin-dependent and -independent mechanisms.[73] Its natriuretic effects, however, seem to be dependent on angiotensin II's conversion to angiotensin III by aminopeptidase N.[74]

Like the AT_1R, the activity of AT_2R may also be modulated by oligomerization, in association with various interacting proteins and ligand-independent effects.[73]

(PRO)RENIN RECEPTOR

In 2002 an apparently novel, 350-amino acid, single-transmembrane protein that binds both renin and prorenin with high affinity was identified.[18] Ligand binding to this protein was shown to induce a fourfold increase in the catalytic cleavage of angiotensinogen as well as stimulating intracellular signaling with activation of MAPKs extracellular signal–regulated kinases 1 and 2 (ERK1/2),[18] leading to it being named the (pro)renin receptor [(P)RR]. The designation (pro)renin refers to its ability to interact with both renin and prorenin.

Given its localization to the mesangium in initial studies, its actions in augmenting local angiotensin II production, and its ability to increase mesangial transforming growth factor-β (TGF-β) production,[75] the (P)RR was understandably implicated in the pathogenesis of kidney disease.[76] However, despite the appeal, it has been difficult to reconcile this view of the (P)RR with a number of other experimental findings, regarding not only its potentially pathogenetic role, but also its pattern of distribution within the kidney and its homology to other proteins. For instance, given the purported pathogenetic role of the (P)RR, the increased abundance of renin that follows the use of ACE inhibitors and ARBs would be expected to be adverse, yet these classes of drugs have been repeatedly shown to be renoprotective. Second, although the (P)RR was initially localized to the glomerular mesangium, more recent detailed studies have shown that (P)RR is primarily expressed in the collecting duct.[77] Third, although initially reported as having no homology with any known membrane protein,[18] database interrogation shows that the (P)RR is identical to two other proteins: CAPER (endoplasmic reticulum–localized type 1 transmembrane adaptor precursor) and ATP6ap2 (ATPase, H^+ transporting, lysosomal accessory protein 2),[78–82] a protein that associates with the vacuolar H^+-ATPase.[83] Indeed, the predominant expression of the (P) RR at the apex of acid secreting cells in the collecting duct, in conjunction with its colocalization and homology with an accessory subunit of the vacuolar H^+-ATPase, suggests that the (P)RR may function primarily in urinary acidification.[77] However, the vacuolar H^+-ATPase is not restricted to the kidney but is widely distributed in the plasma membrane and the membranes of organelles in several tissues where it functions, not only in urinary acidification but also in endocytosis, conversion of proinsulin to insulin, and osteoclast bone resorption.[84] Whereas the prevailing data indicate that (P)RR is an accessory subunit of the vacuolar H^+-ATPase that also binds renin and prorenin, the precise functions of the prorenin- and renin-binding subunit remain to be unraveled in the kidney and elsewhere.

ANGIOTENSIN-CONVERTING ENZYME 2

In 2000, two groups independently reported the existence of the first ACE homolog, ACE2, an apparently novel zinc metalloprotease but with considerable homology (40% identity and 61% similarity) to ACE.[85,86] The gene-encoding ACE2, located on the X chromosome (Xp22), contains 18 exons, several of which bear considerable similarity to the first 17 exons of human *ACE*. Its transcript is 3.4 kb, generating an 805–amino acid peptide that is most highly expressed in kidney, heart, and testis but is also present in plasma and urine.[87,88] In contrast to ACE, ACE2 functions as a carboxypeptidase, removing the terminal phenylalanine from angiotensin II to yield the vasodilatory heptapeptide, angiotensin 1-7 [Ang (1-7)]. ACE2 may also indirectly lead to the formation of Ang (1-7) by cleaving the C-terminal leucine from angiotensin I, thereby generating angiotensin 1-9 [Ang (1-9)] which may then give rise to Ang (1-7) under the influence of ACE or neutral endopeptidase (NEP).[88] Thus ACE2 contributes to both angiotensin II degradation and Ang (1-7) synthesis. Accordingly, ACE and ACE2 were initially viewed as having opposing actions with regard to vascular tone and tissue injury. However, emerging data suggest that the situation is far from clear. For instance, while lentivirus-induced overexpression of ACE2 in the heart exerted a protective influence following experimental myocardial infarction,[89] ACE2 overexpression led to cardiac dysfunction and fibrosis, despite lowering systemic blood pressure.[90]

In the kidney, ACE2 colocalizes with ACE and angiotensin receptors in the proximal tubule while in the glomerulus it is predominantly expressed within podocytes and to a lesser extent in mesangial cells, contrasting the endothelial predilection of ACE at that site.[91] Numerous studies have explored changes in ACE2 expression in human kidney disease as well as in a range of animal models, reporting both increased and decreased levels.[92] As such, it is uncertain whether increased ACE2 might be detrimental or a beneficial response to injury. With this in mind, the findings of intervention studies are of particular importance.

In experimental diabetic nephropathy, for instance, pharmacological ACE2 inhibition with MLN-4760 led to worsening albuminuria and glomerular injury[91]; similar findings were reported in ACE2 knockout mice that were crossed with the Akita model of type 1 diabetes.[93] As might be expected from these findings, augmenting ACE2 activity by the infusion of human recombinant protein (hrACE2) was shown to attenuate diabetic kidney injury in the Akita mouse. In this study, hrACE2 not only improved kidney structure and function but also showed that the protective effects were likely due to reduction in angiotensin II and an increase in Ang (1-7) signaling.[94] In addition to its role in the RAS, ACE2 has been shown to be the receptor for the severe adult respiratory syndrome (SARS) coronavirus.[95] The relevance of some of these findings to the human setting will hopefully be clarified when results of the recently initiated clinical trials of a recombinant human ACE2 peptide (GSK2586881) become available.

ANGIOTENSIN PEPTIDES

Angiotensin III, or Angiotensin-(2-8)

Formed by the actions of aminopeptidase A, the heptapeptide angiotensin III (angiotensin 2-8), like angiotensin II (angiotensin 1-8), exerts its effects by binding to the AT_1R and AT_2R.[96] Initially, angiotensin III was thought to have a predominant role in regulating vasopressin release.[97] However, more recent studies indicate that while angiotensin III is equipotent to angiotensin II with regard to its effects on blood pressure, aldosterone secretion, and renal function, its metabolic clearance rate is approximately five times as rapid.[98]

Angiotensin IV, or Angiotensin-(3–8)

Angiotensin IV is generated from angiotensin III by the actions of aminopeptidase M. Although some of its actions are mediated by the AT_1R, the majority of angiotensin IV's biological effects are thought to result from its binding to insulin-regulated aminopeptidase.[99] Previously viewed as inactive, there has been considerable recent interest in angiotensin IV with regard to its actions in the central nervous system (CNS), where it not only enhances learning and memory but also possesses anticonvulsant properties and protects the brain from ischemic injury.[99]

In addition to its CNS effects, angiotensin IV has also been implicated in atherogenesis, principally related to its ability to activate NF-κB and upregulate several proinflammatory factors that include monocyte chemoattractant protein-1 (MCP-1), intercellular adhesion molecule-1, interleukin-6 (IL-6), and TNF-α, as well as enhancing the synthesis of the prothrombotic factor plasminogen activator inhibitor-1.[100,101] In the kidney, angiotensin IV is reported to have variable effects on blood flow and natriuresis.[99]

Angiotensin-(1-7)

The ostensibly vasodilatory and antitrophic angiotensin 1-7 may be formed by the actions of several endopeptidases that include removal of the terminal tripeptide of angiotensin I by NEP, cleavage of the C-terminal phenylalanine of angiotensin II by ACE2, and the excision of the dipeptidyl group from the C terminal of angiotensin 1-9 by ACE. Evolving evidence indicates that the actions of this heptapeptide are mediated by its binding to the orphan GPCR *masR*.[102] Angiotensin 1-7 induces vasodilation by a number of mechanisms that include the amplification of bradykinin's effects, stimulating cGMP synthesis, and inhibiting the release of norepinephrine.[103] Additionally, angiotensin 1-7 inhibits vascular smooth muscle proliferation and prevents neo-intima formation following balloon injury of the carotid arteries.[104] Contrasting these findings, however, is a recent report that exogenous angiotensin-(1-7), rather than ameliorating diabetic nephropathy, as might have been predicted based on the prevailing paradigm, actually accelerated the progression of the disease.[93]

Angiotensin-(2-10)

In addition to angiotensin II and the other C-terminal cleavage products discussed earlier, angiotensin I (1-10) may also give rise to a number of other potentially biologically active peptides that result from removal of amino acids from its N terminus. Of these, angiotensin 2-10, produced by the actions of aminopeptidase A has been found to modulate the pressor activity of angiotensin II in rodents.[105]

Angiotensin-(1-12)

Angiotensin-(1-12) is a dodecapeptide, first isolated in rat intestine but also found to be present in the kidney and heart, that is cleaved from angiotensinogen by a heretofore unidentified nonrenin enzyme.[106] Notably, in the kidney, angiotensin-(1-12), akin to other components of the intrarenal RAS, is primarily localized to the proximal tubule epithelium.[107] Although its biological activity is incompletely understood, its main mode of action is thought to be mediated by its ability to serve as a precursor to angiotensin II by the site- and possibly species-specific actions of ACE and chymase.[108] Other pathways may, however, also contribute to the overall effects of angiotensin-(1-12), which in the rat kidney may also include the formation of angiotensin-(1-7) and angiotensin-(1-4) by neprilysin.[107]

Angiotensin A and Alamandine

Identified by mass spectroscopy, angiotensin A and alamandine are characterized by the decarboxylation of N-terminal aspartic acid to alanine in angiotensin II and angiotensin-(1-7), respectively.[109] While the extremely low concentrations of angiotensin A suggest that it is unlikely to play a physiological role, this may not be the case for alamandine, which circulates in human plasma and at increased concentrations in patients with end-stage renal disease (ESRD).[109] With its ability to lower blood pressure and reduce fibrosis, the actions of alamandine resemble those of angiotensin-(1-7). However, rather than exerting its effects via the *Mas* receptor, alamandine's actions occur through a related receptor, the *Mas*-related GPCR (*Mrg*D).[110]

INTRARENAL RENIN–ANGIOTENSIN–ALDOSTERONE SYSTEM

In the traditional view of the RAS, angiotensin II functions as a hormone that, in classical endocrine fashion, circulates systemically to act at sites distant from those where it was formed. However, since the cloning of its components, it has become increasingly clear that there is an additional local, tissue-based RAS that functions quasi-independently from its systemic counterpart, acting in paracrine, autocrine, and possibly even intracrine modes.[1] This is most clearly seen in the kidney where pioneering work of Navar and others[111–113] has shown that the kidney possesses all the necessary molecular machinery to synthesize angiotensin II and other bioactive angiotensin peptides. Moreover, their concentrations in glomerular filtrate, tubule fluid, and interstitium are between 10- and 1000-fold higher than in plasma.[38,114]

Within the kidney, renin-expressing cells have traditionally been considered to be terminally differentiated and confined to the JGA. However, in a series of elegant studies using a fate-mapping cre-loxP system, Sequeira López and coworkers showed that renin-expressing cells are precursors to a range of other cell types in the kidney, including those of the arteriolar media, mesangium, Bowman's capsule, and proximal tubule.[115] While normally quiescent, these cells may undergo metaplastic transformation to synthesize renin when homeostasis is challenged.[115] Such threats include not only those related to volume depletion but also tissue injury. For instance, in the setting of single nephron hyperfiltration and consequent progressive dysfunction that follows renal mass reduction, Gilbert and colleagues noted the de novo expression of renin mRNA and angiotensin II peptide in tubule epithelial cells.[116]

In addition to resident kidney cells, infiltrating mast cells may also contribute to activation of the local RAS in disease. Traditionally associated with allergic reactions and host responses to parasite infestation, mast cells have been increasingly recognized for their role in inflammation, immunomodulation, and chronic disease. In the kidney, interstitial mast cell infiltration accompanies most forms of CKD where their abundance correlates with the extent of tubulointerstitial fibrosis and declining GFR, though not proteinuria.[117] Notably, mast cells have been shown to synthesize renin,[118] such that their degranulation will release large quantities of both renin and chymase, accelerating angiotensin II formation in the local environment.

Intracrine Renin–Angiotensin–Aldosterone System

Peptide hormones traditionally bind to their cognate receptors on the plasma membrane and produce their effects through the generation of secondary intermediates. However, emerging evidence suggests that certain peptides may also act directly within the cell's interior, having arrived there by either internalization or intracellular synthesis. For instance, angiotensin II has not only been localized within the cytoplasm and nucleus but its introduction into the cytoplasm was shown more than a decade ago to have major effects on intracellular calcium currents.[119] Uptake of angiotensin II from the extracellular space likely contributes to its intracellular activity; however, recent studies have focused predominantly on its endogenous synthesis. Consistent with the potential role for intracellular angiotensin II, transgenic mice that express an enhanced cyan fluorescent protein–angiotensin II fusion protein that lacks a secretory signal so that it is retained intracellularly develop hypertension with microthrombi in glomerular capillaries and small vessels.[120] To date, numerous canonical and noncanonical pathways in the cytoplasm, nucleus, and mitochondria have been implicated in the intracrine RAS,[121,122] providing a new forefront for the role of the RAS in physiology, pathophysiology, and therapeutics.

RENIN–ANGIOTENSIN–ALDOSTERONE SYSTEM IN KIDNEY PATHOPHYSIOLOGY

In a critical series of experiments in the 1980s, Brenner, Hostetter, and colleagues studied the hemodynamic effects of renal mass ablation in 5/6 nephrectomized rats, a now well-established model of progressive kidney disease.[123] In the setting of nephron loss, those glomeruli that remain undergo compensatory enlargement with increased single nephron GFR (SNGFR) and elevations in intraglomerular pressure (P_{GC}), argued to initiate glomerulosclerosis and loss of function. That this phenomenon might be related to angiotensin II was suggested by previous work in which angiotensin II infusion was demonstrated to also result in elevated P_{GC}.[124] Together, these studies suggested that intraglomerular hypertension, as a consequence of angiotensin II's action, was a pivotal factor underlying the inexorable progression of kidney disease and that strategies to reduce P_{GC} would lead to its amelioration. Indeed, in proof-of-concept studies, blockade of angiotensin II formation with the ACE inhibitor, enalapril, was shown to dilate the glomerular efferent arteriole, reduce P_{GC}, and disease progression in 5/6 nephrectomized rats.[125] By contrast, combination therapy with hydralazine, reserpine, and hydrochlorothiazide, though equally effective in lowering systemic blood pressure, failed to ameliorate intraglomerular hypertension and disease progression.[125] These studies were soon followed by similar ones in other disease models, particularly diabetes, which like the 5/6 nephrectomized rat is also characterized by increased SNGFR and elevated P_{GC}.[126]

FIBROSIS

During the past 20 years considerable research has focused on many of the nonhemodynamic effects of angiotensin II. For instance, in addition to their effects on P_{GC}, ACE inhibitors and ARBs are also highly effective in reducing interstitial fibrosis and tubule atrophy, each close correlates of progressive kidney dysfunction. Underlying these effects is the ability of angiotensin II to potently induce expression of the profibrotic and proapoptotic growth factor and TGF-β in a range of kidney cell types.[127,128] Consistent with these in vitro studies, TGF-β overexpression is seen in both the glomerular and tubulointerstitial compartments in 5/6 nephrectomized rats and diabetic rats, where studies also showed that both ACE inhibitors and ARBs were effective at reducing TGF-β and disease progression.[129,130] Similarly, in human diabetic nephropathy, the ACE inhibitor perindopril was found to reduce TGF-β mRNA in a sequential renal biopsy study[131] and losartan was shown to lower urinary TGF-β excretion.[132]

PROTEINURIA

The development of proteinuria is both a cardinal manifestation of glomerular injury and a pathogenetic factor in the progression of renal dysfunction. While P_{GC} remains an important factor in determining the transglomerular passage of albumin, more recent work has focused on the potential contribution of the podocyte. Indeed, podocyte injury is a cardinal manifestation of proteinuric renal disease where foot process effacement has been shown to be prevented by both ACE inhibition and angiotensin receptor blockade.[133] In consideration of its crucial role in the development and function of the glomerular filtration barrier, other studies have focused on the podocyte slit pore membrane protein nephrin. Of note, podocytes express the AT_1R and respond to the addition of angiotensin II to the cell culture medium by dramatically decreasing their expression of nephrin.[134] Consistent with these findings, the reduction in nephrin expression in patients with diabetic nephropathy was shown to be ameliorated by ACE inhibitor treatment for 2 years.[135]

INFLAMMATION, IMMUNITY, AND THE RENIN–ANGIOTENSIN–ALDOSTERONE SYSTEM

Inflammatory cell infiltration is a long-recognized feature of CKD that is attenuated in rodent models by agents that block the RAS.[136] In the in vitro setting, angiotensin II activates NF-κB by both AT_1R- and AT_2R-dependent pathways, stimulating the expression of a number of potent chemokines such as MCP-1 and regulated on activation, normal T cell expressed and secreted (RANTES) as well as cytokines, like IL-6.[137] In addition to angiotensin II, angiotensin-(1-7), acting via the Mas receptor, activates NF-κB, inducing proinflammatory effects in the kidney under both basal and disease settings.[138]

In addition to macrophages, mast cells, and other components of the innate immune system, the adaptive immune system also appears to be involved in the pathogenesis of angiotensin II–mediated organ injury. Of note, suppression of the adaptive immune system prevents the development of angiotensin II–dependent hypertension in experimental models[139] and adoptive transfer of $CD4^+CD25^+$ regulatory T cells is able to ameliorate angiotensin II–dependent injury.[140]

DIABETES PARADOX

Despite the fact that patients with long-standing diabetes characteristically have low plasma renin,[141–143] suggesting that the RAS is not activated by the disease, agents that block the RAS are the mainstay of therapy in diabetic nephropathy. Compounding this apparent paradox, although PRA is normal or low in diabetes, plasma prorenin is characteristically elevated. This dichotomy suggests differences in cell-specific responses to diabetes since the JGA is the primary source of renin secretion while prorenin is secreted by a much wider range of cell types. In a recent commentary, Peti-Peterdi et al. have ventured to explain the (pro)renin paradox of diabetes.[144] Although early diabetes would lead to augmented succinate and enhanced JGA renin release, elevated angiotensin II levels would thereafter suppress JGA renin secretion. Contrasting this negative feedback at the JGA, angiotensin II has been shown to have the opposite effect in the tubule, with diabetes causing a 3.5-fold increase in collecting duct renin that could be reduced by AT_1R blockade.[145]

ENDOTHELIN

> ***Clinical Relevance***
> The actions of the endothelins (ETs) are mediated by two receptors: ET types A (ET-A) and B (ET-B). ET receptor antagonists (particularly ET-A receptor antagonists) are being investigated for their efficacy in slowing renal decline in chronic kidney diseases, especially diabetic kidney disease. However, whereas ET receptor antagonists lower proteinuria, their development has been hampered by fluid retention and a narrow therapeutic window.

ETs are potent vasoconstrictors that, although expressed primarily in the vascular endothelium, are also notably present within the renal medulla. The biologic effects of the ET system are mediated by two receptors: ET types A (ET-A) and B (ET-B). In the kidneys, these receptors contribute to the regulation of RBF, salt and water balance, and acid–base homeostasis, as well as potentially mediating tissue inflammation and fibrosis. An important therapeutic role has emerged for ET receptor antagonism in the treatment of pulmonary hypertension;[146–149] ET receptor antagonists have been granted regulatory authority approval for this indication in the United States and in Europe.[150] ET receptor blockade as a therapeutic strategy has been investigated in a range of renal diseases. Recent clinical trials have demonstrated the antiproteinuric and antihypertensive properties of ET receptor antagonists, which have a relatively narrow therapeutic window for the treatment of CKD.

STRUCTURE, SYNTHESIS, AND SECRETION OF THE ENDOTHELINS

ETs consist of three 21–amino acid isoforms that are structurally and pharmacologically distinct: ET-1, ET-2, and ET-3. The dominant isoform in the cardiovascular system is ET-1. Differences in the amino acid sequence among the isopeptides are minor. All three isoforms share a common structure with a typical hairpin-loop configuration that results from two disulfide bonds at the amino terminus and a hydrophobic carboxy terminus that contains an aromatic indol side chain at Trp_{21} (Fig. 11.6). Both the carboxy terminus and the two disulfide bonds are responsible for the biologic activity of the peptide. ETs are synthesized from preprohormones by posttranslational proteolytic cleavage mediated by furin and other enzymes. Dibasic pair–specific processing endopeptidases, which recognize Arg-Arg or Lys-Arg paired amino acids, cleave preproETs, reducing their size from approximately 203 to 39 amino acids. Subsequent proteolytic cleavage of the largely biologically inactive big ETs is mediated by endothelin-converting enzymes (ECEs), the key enzymes in the ET biosynthetic pathway. ECE1 and ECE2 are type II membrane-bound metalloproteases whose amino acid sequence is significantly homologous to that of neprilysin (NEP 24.11).

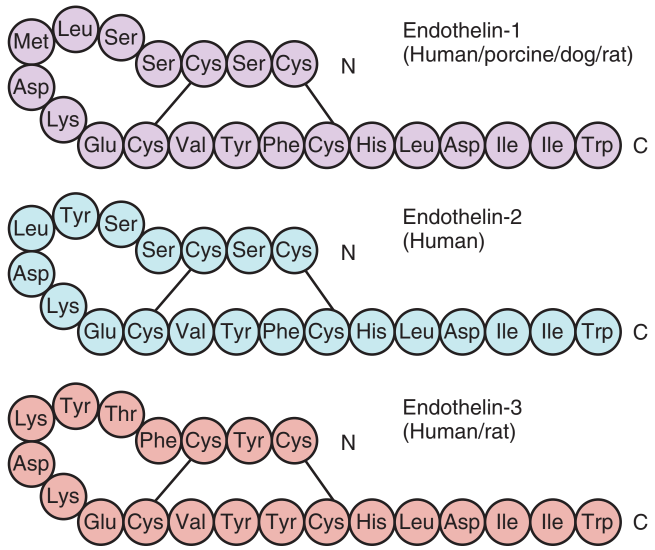

Fig. 11.6 Molecular structure of the three endothelin isoforms. (Modified from Schiffrin EL. Vascular endothelin in hypertension. *Vascul Pharmacol.* 2005;43:19–29.)

Table 11.1 Endothelin Gene and Protein Expression

Stimulation	
Vasoactive Peptides	**Growth Factors**
Angiotensin II	Epidermal growth factor
Bradykinin	Insulin-like growth factor
Vasopressin	Transforming growth factor-β
Endothelin-1	**Coagulation**
Epinephrine	Thromboxane A_2
Insulin	Tissue plasminogen activator
Glucocorticoids	**Other**
Prolactin	Calcium
Inflammatory Mediators	Hypoxia
Endotoxin	Shear stress
Interleukin-1	Phorbol esters
Tumor necrosis factor-α	Oxidized low-density lipoproteins
Interferon-β	
Inhibition	
Atrial natriuretic peptide	Prostacyclin
Brain natriuretic peptide	Protein kinase A activators
Bradykinin	Nitric oxide
Heparin	Angiotensin-converting enzyme inhibitors

Secretion of ET-1 is dependent on de novo protein synthesis, which is constitutive. However, a range of stimuli may also increase ET synthesis through both transcriptional and posttranscriptional regulation (Table 11.1). Once it is synthesized, ET-1 is secreted by endothelial cells into the basolateral compartment, toward the adjacent smooth muscle cells. Because of its abluminal secretion, plasma levels of ET-1 do not necessarily reflect its production.[151]

Within the kidneys, ET-1 expression is most abundant in the inner medulla. In fact, this region possesses the highest concentration of ET-1 of any tissue bed.[152] In addition to their presence in the inner medullary collecting ducts (IMCDs), ETs have also been described in glomerular endothelial cells,[153] glomerular epithelial cells,[154] mesangial cells,[155] vasa recta,[156] and tubule epithelial cells.[157] The kidneys also synthesize ET-2 and ET-3, although at much lower levels than they do ET-1.[158] As with ET-1, ECE1 mRNA is also more abundant in the renal medulla than in the cortex under normal conditions. However, in disease states such as chronic heart failure, ECE1 mRNA is upregulated primarily within the cortex.[159] In human kidneys, ECE1 has been localized to endothelial cells and tubule epithelial cells in the cortex and medulla.[160]

ENDOTHELIN RECEPTORS

ETs bind to two seven-transmembrane domain GPCRs, ET-A and ET-B. Within the vasculature, ET-A receptors are found on smooth muscle cells, where they mediate vasoconstriction. Although ET-B receptors localized on vascular smooth muscle cells can also mediate vasoconstriction, they are also expressed on endothelial cells, where their activation results in vasodilation through the production of NO and prostacyclin.[161] In addition to their role in mediating vascular tone, ET-B receptors also act as clearance receptors for ET-1,[162] particularly in the lung, where ET-B receptor binding accounts for approximately 80% of clearance.[163] Because of its natriuretic and vasodilatory actions, the ET-B receptor is generally considered to confer predominantly renoprotective effects.

In the kidneys, expression of both ET-A and ET-B receptors is most prominent within the IMCDs, although binding of ET-1 also occurs in smooth muscle cells, endothelial cells, renomedullary interstitial cells, thin descending limbs, and medullary thick ascending limbs.[156] ET-A receptors are localized to several renovascular structures, including vascular smooth muscle cells, arcuate arteries, and pericytes of descending vasa recta, as well as glomeruli. ET-B receptors, although prominently represented within the medullary collecting system, have also been demonstrated in proximal convoluted tubules, collecting ducts of the inner cortex, medullary thick ascending limbs, and podocytes.[164]

PHYSIOLOGIC ACTIONS OF ENDOTHELIN IN THE KIDNEY

The ETs have several effects on normal renal function, including regulation of RBF, sodium and water balance, and acid–base homeostasis. Although ET-1 has hemodynamic effects in almost all vessels, the sensitivity of different vascular beds varies. The renal vasculature, along with the mesenteric vessels, is the most sensitive: vasoconstriction occurs at picomolar concentrations of ET-1,[165,166] increasing RVR and decreasing RBF. However, long-lasting vasoconstriction that is mediated by the ET-A receptor may be preceded by a transient ET-B receptor–mediated vasodilation.[167] Because of the site-specific distribution of ET receptors, ET-1 may exert different vasoconstrictive and vasodilatory effects in different regions of the kidneys. For example, by inducing NO release from adjacent tubule epithelial cells, ET-1 may actually increase blood flow in the renal medulla, where ET-B receptors predominate.[168]

In addition to effects on RBF, the ET system also plays a direct role in renal sodium and water handling.[169] In the renal medulla, ET is regulated by sodium intake and exerts its natriuretic and diuretic effects through the ET-B receptor.[170–172] In addition to natriuretic and diuretic effects, the ET-B receptor may also contribute to acid–base homeostasis by stimulating proximal tubule sodium/proton exchanger isoform 3 (NHE3).[173] Although the role of ET-B receptor activation in urinary sodium excretion has been appreciated for some time, more recent evidence suggests that renal medullary ET-A receptors may also mediate natriuresis.[174] This may partly explain the edema that can occur as a side effect of ET-A or dual ET receptor antagonism.

ROLE OF ENDOTHELIN IN ESSENTIAL HYPERTENSION

In view of its potent vasoconstrictive properties, it is not surprising that ET-1 has been implicated in the pathogenesis of hypertension. In preclinical models of hypertension, ET antagonism may ameliorate heart failure, vascular injury, and renal failure, as well as reduce the incidence of stroke.[175,176] ET-A receptor antagonism has also been shown to normalize blood pressure in rats exposed to eucapnic intermittent hypoxia, which is analogous to sleep apnea in humans.[177]

PreproET-1 mRNA is increased in the endothelium of subcutaneous resistance arteries in patients with moderate to severe hypertension[178] and according to a recent meta-analysis, plasma ET-1 concentrations are increased in individuals with hypertension.[179] However, plasma ET-1 levels are not universally elevated[175]; an increase is found more commonly in the presence of end-organ damage or in salt-depleted, salt-sensitive patients with a blunted renin response.[180] A major component of this increase in disease is often decreased clearance by the kidneys. These findings suggest that certain patient subgroups may be more responsive than others to ET receptor blockade. Females appear to be relatively protected from the pressor effects of ET-1 by virtue of both increased ET-B expression and a blunted hemodynamic response to ET-A receptor activation.[181]

Clinical trials of the antihypertensive effects of ET receptor antagonism have been hampered by difficulties with selectivity for the ET-A receptor, study design, dosing regimens, and adverse events.[182] Because the ET-B receptor exerts diuretic and natriuretic effects, induces vasodilation, and clears ET-1, selective ET-A receptor antagonists may be expected to demonstrate a more favorable antihypertensive profile.[182] Mixed ET-A/B and specific ET-A receptor antagonists are distinguished by their in vitro binding affinities with mixed ET-A/B receptor antagonists demonstrating a selectivity for ET-A of <100-fold, and ET-A selective antagonists having an affinity for the ET-A receptor of 100-fold or higher. However, it has been suggested that a 1000-fold or higher affinity may be required in order to induce ET-A receptor-specific effects in vivo.[183,184] In an early study, treatment of patients with essential hypertension with the nonselective ET receptor antagonist bosentan decreased blood pressure as effectively as enalapril, without reflex neurohumoral activation, over a 4-week period.[185] Similarly, in 115 patients with resistant hypertension who were taking three or more agents, the selective ET-A receptor antagonist darusentan significantly reduced blood pressure at 10 weeks.[186] In a subsequent study of 379 individuals with resistant hypertension, darusentan treatment for 14 weeks reduced blood pressure by approximately 18/10 mm Hg with no evidence of dose dependence across a range of 50 to 100 mg/day.[187] However, in a second study of similar design a large placebo effect meant that darusentan treatment failed to achieve its primary endpoint of change in office blood pressure and the development of the drug for this indication was halted.[188] Interestingly, in both of these studies ambulatory blood pressure monitoring revealed a reduction in systolic blood pressure with active treatment.[183,189] However, also in both studies, peripheral edema or fluid retention was more common in patients treated with darusentan than those receiving placebo.[187,188]

ROLE OF ENDOTHELIN IN RENAL INJURY

Beyond its effects on the regulation of vascular tone, the ET system also likely plays a direct role in the pathogenesis of fibrotic injury in CKD.[190] In patients with CKD, plasma ET-1 concentrations are elevated, as a result of both increased production and decreased renal clearance.[191] Urinary levels of ET-1 are also increased, which is indicative of increased renal ET-1 expression.[192] One mechanism for increased renal ET-1 in CKD is a direct effect of urinary protein on ET-1 expression in tubule epithelial cells.[193,194] Beyond direct effects of urine protein, a number of proinflammatory factors induce ET-1 expression in the kidneys, including hypoxia, angiotensin II, thrombin, thromboxane A_2, TGF-β, and shear stress (Table 11.1).

Several distinct mechanisms may account for the injurious effects of ET-1 on the kidneys. Locally derived ET-1 has direct hemodynamic effects, increasing P_{GC} at high doses and causing vasoconstriction of the vasa recta and peritubular capillaries, with a resultant reduction in tissue oxygen tension. ET-1 acts as a chemoattractant for inflammatory cells, which may express the peptide themselves, stimulating interstitial fibroblast and mesangial cell proliferation and mediating the production of a number of factors associated with collagenous matrix deposition, including TGF-β, matrix metalloproteinase-1, and tissue inhibitors of metalloproteinases-1 and -2. In mesangial cells, ET-1 can induce cytoskeletal remodeling and cell contraction[195] and, in these cells, ET-A receptor activation appears to be important to the development of Alport glomerular disease.[196]

A particularly important role of ET-1 in mediating injury of glomerular podocytes is also beginning to emerge. For instance, increased passage of protein across the filtration barrier causes podocyte cytoskeletal rearrangements and coincident upregulation of ET-1, which may act in an autocrine manner to further propagate ultrastructural injury in the same cells.[197–199] ET-1 promotes podocyte dedifferentiation and migration by activating ET-A and increasing β-arrestin-1 expression which ultimately results in EGFR transactivation, phosphorylation of β-catenin, and increased expression of the transcription factor Snail, an inducer of epithelial–mesenchymal transition.[200] ET-1 induces calcium signaling in podocytes through both the ET-A and ET-B receptors, and mice with deletion of both the ET-A receptor and ET-B receptor from podocytes are protected from the glomerular injury associated with streptozotocin-induced diabetes.[201] Using these mice, investigators have subsequently gone on to demonstrate that ET-1 causes podocytes to release heparanase that damages the endothelial glycocalyx facilitating albumin passage across the filtration barrier.[202]

THE ENDOTHELIN SYSTEM IN CHRONIC KIDNEY DISEASE AND DIABETIC NEPHROPATHY

PRECLINICAL STUDIES OF ET RECEPTOR ANTAGONISTS IN DIABETIC KIDNEY DISEASE

ET receptor antagonists have been employed to study the role of ETs in renal pathophysiology in a range of experimental models, including the rat remnant kidney, lupus nephritis, and diabetes. In the remnant kidney model of progressive renal disease, although beneficial effects have been reported with nonselective ET receptor antagonists,[203] selective ET-A receptor inhibition appears to yield superior outcomes with concomitant inhibition of ET-B receptors, potentially abrogating any beneficial effects.[204]

Data with regard to an effect of high glucose concentrations on ET synthesis and secretion are conflicting. Mesangial cell p38 MAPK activation in response to ET-1, angiotensin II, and PDGF is enhanced in the presence of high glucose levels.[205] By contrast, mesangial contraction in response to ET-1 is diminished under high-glucose conditions.[206,207] Circulating ET-1 concentrations are elevated in animal models of both type 1 and type 2 diabetes. Increased expression of

ET-1 and its receptors has been found in glomeruli and in tubule epithelial cells,[194,208] although increased expression of ET receptors has not been a universal finding.[209] Diabetes also causes an increase in renal ECE1 expression, the effect being synergistic with that of radiocontrast media.[210]

A number of researchers have investigated the effect of both nonselective and selective ET-A receptor antagonists in experimental diabetic nephropathy. In streptozotocin-diabetic rats, the nonselective ET receptor antagonist bosentan has yielded conflicting results,[211,212] whereas another nonselective ET receptor antagonist, PD142893, improved renal function when administered to streptozotocin-diabetic rats that were already proteinuric.[194] More recently, acute ET-A receptor antagonism was shown to improve oxygen availability in the kidneys of streptozotocin-diabetic rats.[213] In Otsuka Long Evans Tokushima Fatty (OLETF) rats with type 2 diabetes, selective ET-A receptor blockade attenuated albuminuria, without affecting blood pressure, whereas ET-B receptor blockade had no effect.[214] In a study of streptozotocin-diabetic apolipoprotein E knockout mice, the renoprotective effects of the predominant ET-A receptor antagonist avosentan were comparable or superior to the ACE inhibitor quinapril.[215] Supporting a protective role for the ET-B receptor, diabetic ET-B receptor–deficient rats developed severe hypertension and progressive renal failure.[216]

Accumulation of reactive oxygen species plays a major role in the pathogenesis of diabetic complications, particularly diabetic nephropathy,[217,218] and several observations suggest that the ET system may contribute to oxidative stress. In low-renin hypertension, ET-1 increases superoxide in carotid arteries,[219] and ET-A receptor blockade decreases vascular superoxide generation.[220,221] Similarly, ET-1 infusion increased urinary excretion of 8-isoprostane prostaglandin $F_2\alpha$ in rats, which is indicative of increased generation of reactive oxygen species.[222] By contrast, however, other preclinical studies have suggested a predominantly proinflammatory role for ET-1 in diabetic nephropathy. For instance, the selective ET-A receptor antagonist ABT-627 prevented the development of albuminuria in streptozotocin-diabetic rats without an improvement in markers of oxidative stress but with a reduction in macrophage infiltration and urinary excretion of TGF-β and prostaglandin E_2 metabolites.[223]

CLINICAL STUDIES OF ET RECEPTOR ANTAGONISTS IN CKD AND DIABETIC NEPHROPATHY

Both plasma and urinary ET-1 levels are increased in patients with CKD[191,224,225] with plasma ET-1 levels inversely correlating with estimated GFR. In a study of hypertensive patients with CKD, both selective ET-A receptor blockade and nonselective ET receptor blockade lowered blood pressure.[226] However, ET-A receptor blockade increased both RBF and effective filtration fraction and decreased RVR, whereas dual blockade had no effect.[226]

In a study of 22 nondiabetic individuals with CKD, intravenous infusion of the ET-A receptor antagonist BQ-123 reduced pulse wave velocity and proteinuria to a greater extent than the calcium channel antagonist nifedipine which comparably lowered blood pressure.[227] These findings suggest a potentially blood pressure–independent mechanism of action for the antiproteinuric effect observed. In a subsequent study by the same investigators, 27 subjects with proteinuric CKD were treated with the ET-A receptor antagonist sitaxsentan for 6 weeks in a three-way crossover study design. Sitaxsentan treatment was associated with a reduction in blood pressure, proteinuria, and pulse-wave velocity, whereas nifedipine reduced pulse-wave velocity and blood pressure but had no effect on urine protein excretion.[228] Subsequently, sitaxsentan was also shown to restore the nocturnal dip in blood pressure in people with CKD.[229] A fall in GFR with sitaxsentan therapy observed in this study is analogous to that seen with RAS blockade.[228] Although no clinically significant adverse effects were seen, sitaxsentan development has subsequently been halted due to hepatotoxicity.

The effect of the ET-A receptor antagonist avosentan was examined, in addition to standard treatment with an ACE inhibitor or ARB, in a placebo-controlled trial of 286 patients with diabetic nephropathy and macroalbuminuria.[230] At 12 weeks, avosentan was found to decrease urine albumin excretion rate without affecting blood pressure. These results led to the initiation of the ASCEND trial (a randomised, double blind, placebo controlled, parallel group study to assess the effect of the endothelin receptor antagonist avosentan on time to doubling of serum creatinine, end-stage renal disease or death in patients with type 2 diabetes mellitus and diabetic nephropathy).[231] ASCEND set out to examine the effects of avosentan, on top of RAS blockade, in 1392 individuals with type 2 diabetes and nephropathy but was terminated after a median duration of 4 months due to adverse events.[231] In that study, despite a more than 40% reduction in urine albumin-to-creatinine ratio with avosentan, adverse events, predominantly fluid overload and congestive heart failure, occurred more frequently in those receiving active therapy than placebo.[231]

Since the publication of ASCEND, it has become increasingly apparent that the outcome of that study was hampered by the relatively high dose of avosentan that was selected and that ET-A receptor antagonists likely have a relatively narrow therapeutic window. However, if this therapeutic window is appropriately targeted and patients carefully selected, then ET-A receptor antagonists may still find a clinical niche for the treatment of diabetic nephropathy. The Reducing Residual Albuminuria in Subjects with Diabetes and Nephropathy with Atrasentan trial and an identical study conducted in Japan (RADAR/JAPAN) explored the effects of two different doses of the ET-A receptor antagonist atrasentan (0.75 and 1.25 mg/day, respectively) in 211 patients with type 2 diabetes and albuminuria (estimated GFR [eGFR] 30–75 mL/min/1.73 m^2).[232] In comparison with placebo, 0.75 and 1.25 mg/day atrasentan, on top of maximum tolerated doses of ACE inhibitor or ARB, reduced urine albumin-to-creatinine ratio by an average of 35% and 38%, respectively, without major side effects.[232] Atrasentan treatment was also associated with decreases in 24-hour blood pressure, low-density lipoprotein cholesterol, and triglycerides.[232] Importantly, despite a comparable lowering of albuminuria to the 0.75 mg/day dose of atrasentan, the 1.25 mg/day atrasentan dose was accompanied by more fluid retention.[232] In a post hoc analysis, fluid retention was more likely in participants with lower eGFR and a higher dose of atrasentan, whereas the degree of urinary albumin lowering was not linked to the degree of fluid retention.[233] Thus it is likely that the antiproteinuric and fluid retaining effects of ET receptor antagonists are mediated by different mechanisms; plausibly the former by vascular or glomerular actions, and the latter by direct effects of ET receptor blockade on sodium transport in the renal tubule.[233] Regardless, based on the promising results of RADAR/JAPAN, the 0.75 mg/day dose of atrasentan is currently being investigated in the phase 3 Study of Diabetic Nephropathy with Atrasentan (SONAR) trial, which has a primary outcome of hard renal endpoints (time to doubling of serum creatinine or ESRD, including renal death; NCT01858532). This study aims to recruit over 4000 participants and is estimated to complete at the end of 2018.

COMBINED ECE AND NEPRILYSIN INHIBITION

Distinct from ET receptor antagonism, blockade of ET-1-induced signaling has been explored in the clinical setting with the use of the combined ECE and neprilysin inhibitor daglutril.[234] In an 8-week, crossover design study of participants with type 2 diabetes, blood pressure <140/90 mm Hg, and urinary albumin excretion 20–999 μg/min, daglutril (300 mg/day) did not significantly reduce albuminuria compared with placebo, although blood pressure was reduced.[234] The failure to reach the primary endpoint of albuminuria reduction may relate to concurrent neprilysin inhibition, which may diminish ET-1 degradation.[235] Alternatively, it may reflect an overall diminution of ET-1 with consequent decreased activation of ET-B as well as ET-A.[236]

THE ENDOTHELIN SYSTEM AND OTHER KIDNEY DISEASES

In addition to diabetic and nondiabetic CKD, the role of the ET system has also been investigated in a number of other kidney diseases. Overall, these studies have suggested some degree of renoprotection with either selective ET-A or nonselective ET receptor inhibitors.

SICKLE CELL DISEASE-ASSOCIATED NEPHROPATHY

Administration of the selective ET-A receptor antagonist ambrisentan preserved GFR and prevented the development of albuminuria in a humanized mouse model of sickle cell disease (SCD).[237] However, the renoprotective effects of ambrisentan were only partially recapitulated by treatment with the combined ET-A/ET-B receptor antagonist A-182086, highlighting the importance of selectively targeting the ET-A receptor in SCD.[237]

RENOVASCULAR DISEASE

A series of recent studies in pigs provide support for ET-A receptor blockade in the treatment of renovascular disease. In an initial study, investigators treated pigs with unilateral renal artery stenosis with an ET-A receptor antagonist beginning at the onset of renovascular disease and continuing for 6 weeks.[238] In these experiments, researchers observed that ET-A receptor blockade preserved renal hemodynamics, renal function, and microvascular architecture in the stenotic kidney.[238] Similarly, ET-A receptor blockade (but not ET-B receptor blockade) reversed microvascular rarefaction and diminished renal inflammation and fibrosis when it was initiated 6 weeks after the induction of renal artery stenosis.[239] Finally, ET-A receptor blockade also led to an improvement in microvascular density and renal function recovery compared with placebo when it was administered following percutaneous transluminal renal angioplasty/stenting.[240]

ACUTE KIDNEY INJURY

ET-1 may play a role in sepsis-mediated acute renal failure,[241] although experimental findings have been conflicting, dependent to some extent on the ET receptor antagonist employed. For example, in a rat model of early normotensive endotoxemia, neither an ET-A receptor antagonist nor a combined ET-A/ET-B receptor blockade improved GFR,[242] whereas ET-B receptor blockade alone resulted in a marked reduction in RBF.[242] By contrast, in a porcine model of endotoxemic shock, the dual ET receptor antagonist tezosentan attenuated the decrease in RBF and increase in plasma creatinine.[243] Pointing to a role of ET-1/ET-A signaling in the progression from acute kidney injury to CKD, transient unilateral renal ischemia induced upregulation of ET-1 and ET-A receptor in mice and ET-A receptor antagonism (but not ET-B receptor antagonism) prevented progressive kidney injury.[244]

SYSTEMIC LUPUS ERYTHEMATOSUS

Urinary ET-1 excretion is correlated with disease activity in patients with systemic lupus erythematosus (SLE),[245] and serum from such patients has been shown to stimulate ET-1 release from endothelial cells in culture.[246] In accordance with a pathogenetic role for the ET system in SLE, the ET-A receptor antagonist FR139317 attenuated renal injury in a murine model of lupus nephritis.[247]

PRIMARY FOCAL AND SEGMENTAL GLOMERULOSCLEROSIS

Sparsentan is a dual ET-A receptor/ARB antagonist being developed for the treatment of primary focal and segmental glomerulosclerosis (FSGS) by Retrophin Inc. In late 2016, the company reported results from the phase 2 DUET study that examined the effect of three different doses of sparsentan (200, 400, and 800 mg/day) when compared with the ARB irbesartan (300 mg/day) in 96 participants over an 8-week period. The mean reduction in proteinuria in sparsentan-treated patients was 45% in comparison to a 19% reduction in those receiving irbesartan.[248]

SCLERODERMA

The Zibotentan Better Renal Scleroderma Outcome Study (ZEBRA) is a 3-part phase 2 study (ZEBRA 1, ZEBRA 2A, and ZEBRA 2B) exploring the safety and therapeutic potential of the ET-A receptor antagonist zibotentan in acute and chronic renal complications of scleroderma (NCT02047708). The primary outcome measure is the plasma level of soluble vascular cell adhesion molecule-1 as a biomarker of scleroderma renal involvement.

HEPATORENAL SYNDROME

Plasma ET-1 concentrations are increased in individuals with cirrhosis and ascites and in patients with type 2 hepatorenal syndrome (diuretic-resistant or refractory ascites with slowly progressive renal decline) in whom systemic vasodilation accompanies paradoxical renal vasoconstriction.[249] To investigate the therapeutic potential of ET receptor antagonism in this setting, the combined ET-A/ET-B receptor blocker tezosentan was administered to six patients in an early phase clinical trial.[250] In this study, treatment was discontinued early in five patients, in one case because of systemic hypotension and in four because of concerns about worsening renal function.[250] These adverse effects are consistent with a dose-dependent decline in renal function in patients with acute heart failure treated with tezosentan, and they highlight the need for caution with the use of ET receptor antagonists in certain patient populations.

PREECLAMPSIA

A role for ET-1 in the development of preeclampsia is suggested by the observations that infusion of fms-like tyrosine kinase-1 and TNF-α into pregnant rats induced ET-A-dependent hypertension,[251–253] whereas ET-A receptor antagonism attenuated placental ischemia-induced hypertension in a rat model.[254] Despite the mechanistic role of ET-1 in the pathogenesis of preeclampsia, however, ET receptor antagonists are very unlikely to be used in this condition given their known teratogenicity.[251]

SAFETY PROFILE OF ENDOTHELIN RECEPTOR ANTAGONISTS

The therapeutic development of ET receptor antagonists has been slowed by the adverse side effect profile of available agents, particularly the dose dependency of certain adverse effects. Most notable has been the development of fluid retention, peripheral edema, and congestive heart failure despite the use of predominant ET-A receptor antagonists. The mechanisms that underlie the fluid retention associated with ET-A receptor antagonism have not been fully resolved. It has been suggested that the use of comparatively high doses of ET-A receptor antagonists may have resulted in concurrent ET-B receptor blockade. However, inhibition of nephron ET-A receptors may also be implicated.[167,255] For instance, mice with nephron or collecting duct ET-A receptor deletion were protected from the fluid retention associated with ET-A receptor blockade.[256] Hepatotoxicity may be a class effect or may be restricted to particular subclasses of ET receptor antagonist. A rise in hepatic transaminases has been observed with both bosentan and sitaxsentan, which are both sulfonamide-based agents, but not with ambrisentan or darusentan, which are propionic acid based.[183,186,187,257,258] As discussed earlier, teratogenicity would preclude the use of this class of agents in pregnancy, whereas the potential for testicular toxicity has also been described, although testicular damage has not been reported in patients taking ET receptor antagonists for the treatment of pulmonary hypertension.[236]

NATRIURETIC PEPTIDES

Clinical Relevance
Neprilysin inhibitors prevent the enzymatic degradation of the natriuretic peptides. When used alone they do not produce sustained antihypertensive effects, likely a consequence of compensatory upregulation of the renin–angiotensin system. The combination of neprilysin inhibition and angiotensin-converting enzyme inhibition is associated with an increased risk of angioedema. The combination of angiotensin receptor blockade and an inhibitor of neprilysin has a more favorable side-effect profile and has demonstrated efficacy in the treatment of heart failure. The effect of combination angiotensin receptor blockade/neprilysin inhibition on hard renal outcomes is currently unknown.

The NPs are a family of vasoactive hormones that play a role in salt and water homeostasis. The family consists of at least five structurally related peptides: ANP, BNP, CNP, *Dendroaspis* natriuretic peptide (DNP), and urodilatin. ANP was originally isolated from human and rat atrial tissues in 1984.[259] Since then, the NP family has been found to include several other members, all of which share a common 17–amino acid ring structure that is stabilized by a cysteine bridge and that contains several invariant amino acids.[260] Both BNP[261] and CNP[262] were originally identified in porcine brain tissue, and DNP was first isolated from the venom of the green mamba snake *Dendroaspis angusticeps*.[263] Urodilatin is an NH2-terminally extended form of ANP that was initially described in human urine.[264] NP inactivation occurs through at least two distinct pathways: binding to a clearance receptor (natriuretic peptide receptor [NPR]-C) and enzymatic degradation. Other peptides that may be involved in salt and water balance include guanylin, uroguanylin, and adrenomedullin.

ANP and BNP act as endogenous antagonists of the RAS-mediating natriuresis, diuresis, vasodilation, and suppression of sympathetic activity, as well as inhibiting cell growth and decreasing secretion of aldosterone and renin.[265] The role of NPs in cardiovascular and renal disease, particularly BNP, has led to their adoption into clinical practice as indicators of disease states and, to some extent, as therapeutic agents.

STRUCTURE AND SYNTHESIS OF THE NATRIURETIC PEPTIDES

ATRIAL NATRIURETIC PEPTIDE

ANP is a 28–amino acid peptide comprising a 17–amino acid ring linked by a disulfide bond between two cysteine residues and a COOH-terminal extension that confers its biologic activity (Fig. 11.7). The gene for ANP, *NPPA,* is found on chromosome 1p36 and encodes the precursor preproANP, which is between 149 and 153 amino acids in length according to the species of origin. Human preproANP consists of 151 amino acids and is rapidly processed to the 126–amino acid proANP. ANP is identical in mammalian species except for a single amino acid substitution at residue 110, which is isoleucine in rat, rabbit, and mouse and methionine in human, pig, dog, sheep, and cow.

ANP synthesis occurs primarily within atrial cardiomyocytes, in which it is stored as proANP, the main constituent of the atrial secretory granules. The major stimulus to ANP release is mechanical stretch of the atria that is secondary to increased wall tension. In addition to atrial stretch, ANP synthesis and release may be stimulated by neurohumoral factors such as glucocorticoids, ET, vasopressin, and angiotensin II, partly through changes in atrial pressure and partly through direct cellular effects. Although ANP mRNA levels are approximately 30- to 50-fold higher in the cardiac atria than in the ventricles, ventricular expression is dramatically increased in the developing heart and in conditions of hemodynamic overload such as heart failure and hypertension. Beyond the heart, ANP has also been demonstrated in the kidneys, brain, lungs, adrenal glands, and liver. In the kidneys, alternate processing of proANP adds four amino acids to the NH_2 terminus of the ANP peptide to generate a 32–amino acid peptide: proANP 95-126, or urodilatin.

ANP is stored, primarily as proANP, in the secretory granules of the atrial cardiomyocytes and is released by fusion of the granules with the cell surface. During this process, proANP is cleaved to an NH_2-terminal 98–amino acid peptide (ANP

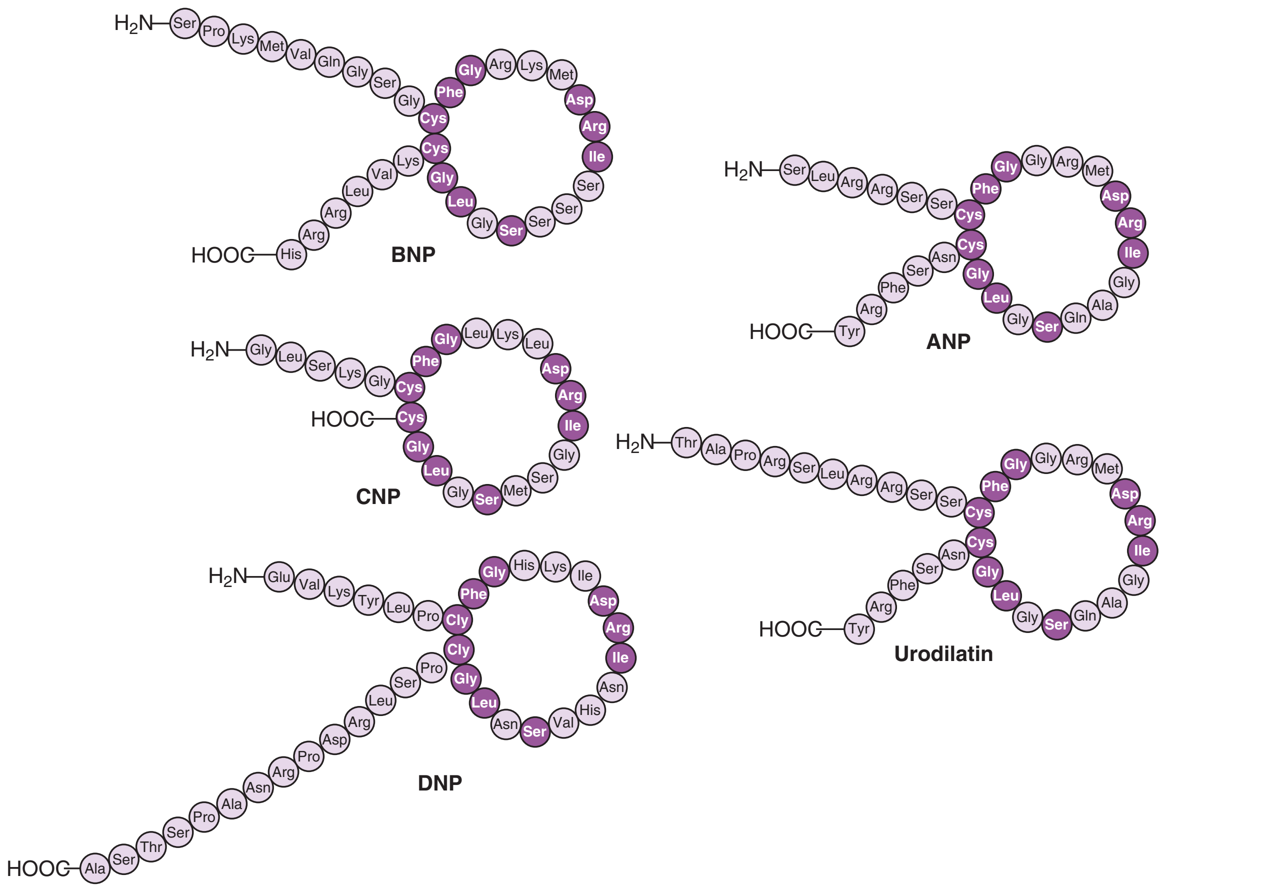

Fig. 11.7 Molecular structure of the natriuretic peptides. *ANP,* Atrial natriuretic peptide; *BNP,* brain natriuretic peptide; *CNP,* C-type natriuretic peptide; *DNP, Dendroaspis* natriuretic peptide. (Modified from Cea LB. Natriuretic peptide family: new aspects. *Curr Med Chem Cardiovasc Hematol Agents.* 2005;3:87–98.)

1-98) and the COOH-terminal 28–amino acid biologically active fragment (ANP 99-126). Both fragments circulate in the plasma; further processing of the NH_2-terminal fragment leads to the generation of peptides ANP 1-30 (long-acting NP), ANP 31-67 (vessel dilator), and ANP 79-98 (kaliuretic peptide), all of whose biologic actions may be similar to those of ANP.[266]

BRAIN NATRIURETIC PEPTIDE

The BNP gene, NPPB, is located only about 8 kb upstream of the ANP gene on the short arm of chromosome 1 in humans, which suggests that the two genes may share both evolutionary origin and transcriptional regulation. By contrast, NPPC, the gene encoding CNP, is found separately, on chromosome 2. CNP is highly conserved across species; thus it may represent the evolutionary ancestor of ANP and BNP. BNP, like ANP, is synthesized as a preprohormone, between 121 and 134 amino acids in length, according to species of origin. Human preproBNP (134 amino acids) is cleaved to produce the 108–amino acid precursor proBNP. Further processing leads to the production of the 32–amino acid, biologically active BNP (which corresponds to the C terminal of the precursor), as well as a 76–amino acid N-terminal fragment (NT-proBNP).[267] Active BNP, NT-proBNP, and pro-BNP all circulate in the plasma. Circulating BNP contains the characteristic 17–amino acid ring structure closed by a disulfide bond between two cysteine residues, along with a nine–amino acid N-terminal tail and a six–amino acid C-terminal tail (Fig. 11.7).[268]

The term *brain natriuretic peptide* is somewhat misleading, given that the primary sites of synthesis of BNP are the cardiac ventricles, and expression also occurs, to a lesser extent, in atrial cardiomyocytes. Like ANP, expression of BNP is regulated by changes in intracardiac pressure and stretch. However, unlike ANP, which is stored and released from secretory granules, BNP is regulated at the gene expression level and is synthesized and secreted in bursts. BNP expression is increased in heart failure, hypertension, and renal failure. Its plasma half-life is approximately 22 minutes; by contrast, the half-life of circulating ANP is 3 to 5 minutes, and the half-life of the biologically inactive NT-proBNP is 120 minutes. This difference is relevant to the utility of NP measurement as a biologic marker of cardiorenal disease. Changes in pulmonary capillary wedge pressure may be reflected by plasma BNP concentrations every 2 hours and by NT-proBNP levels every 12 hours.[269,270] The physiologic actions of BNP are similar to those of ANP, including effects on the kidneys (natriuresis and diuresis), vasculature (hypotension), endocrine system (inhibition of plasma renin and aldosterone secretion), and the brain (central vasodepressor activity).

C-TYPE NATRIURETIC PEPTIDE

As is the case for ANP and BNP, CNP is derived from a prepropeptide that undergoes posttranslational proteolytic cleavage. The initial translation product preproCNP is 126 amino acids in length and is cleaved to produce the 103–amino acid prohormone. Cleavage of proCNP yields two mature peptides made up of 22 and 53 amino acids: CNP and NH_2-terminally extended form of CNP, respectively. Of the 17 amino acids within the CNP ring structure, 11 are identical to those in the other NPs, although, uniquely, CNP lacks an amino tail at the carboxy terminus (Fig. 11.7). Whereas ANP and BNP are ligands for a guanylyl cyclase–coupled receptor, the NPR-A receptor, CNP is a specific ligand for the NPR-B receptor. CNP primarily functions in an autocrine/paracrine manner with effects on vascular tone and muscle cell growth.[271] Expression of the CNP gene by the endothelial cells, the presence of CNP receptors on vascular smooth muscle cell, and the antiproliferative effect of CNP on vascular smooth muscle cells suggest that CNP is produced by the endothelium and acts on adjacent cells, serving as an autocrine/paracrine endothelium-derived locally active vasoregulatory system. Accordingly, plasma concentrations of CNP are very low, although they are increased in the conditions of heart failure and renal failure. CNP is present in the heart, kidneys, and endothelium, and its receptor is also expressed in abundance in the hypothalamus and pituitary gland, which suggests that the peptide may also play a role as a neuromodulator or neurotransmitter. Regulation of CNP expression is distinct from that of ANP and BNP and is controlled by a number of vasoactive mediators, including insulin, vascular endothelial growth factor, TGF-β, TNF-α, and IL-1β.[271]

The principle enzymes responsible for the conversion of proANP, proBNP, and proCNP to their active forms are the serine proteases, corin and furin.[272] Corin converts proANP to ANP,[273] furin converts proCNP to CNP[274] and both corin and furin cleave proBNP.[275] Corin is highly expressed in the heart and to a lesser extent in the kidney[272] and it is the rate-limiting enzyme in ANP activation.[276] In response to pressure overload, corin-deficient mice develop hypertension together with cardiac hypertrophy and dysfunction.[277,278] In the kidneys, corin colocalizes with ANP[279] and decreased urinary corin excretion has been observed in patients with CKD.[280] Interestingly, studies combining observations made in organ-specific corin-deficient mice together with human correlative experiments have identified a role for impaired uterine corin/ANP function in the pathogenesis of preeclampsia.[281]

DENDROASPIS NATRIURETIC PEPTIDE

The physiologic role of DNP has been controversial since its original identification in the venom of the Green Mamba snake, *D. angusticeps,* in 1992.[263,282] DNP is a 38–amino acid peptide that shares the 17–amino acid ring structure common to all NPs, except that it has unique N- and C-terminal regions (Fig. 11.7).[270] Immunoreactivity for DNP has been reported in human plasma and atrial myocardium, and DNP has also been described in rat[283] and rabbit[284] kidneys, rat colon,[285] rat aortic vascular smooth muscle cells,[286] and pig ovarian granulosa cells.[287] DNP binds to NPR-A[288] and the clearance receptor NPR-C,[289] which may be of particular relevance in view of the peptide's apparent resistance to enzymatic degradation.[290] In dogs, either under normal conditions or in a pacing-induced heart failure model, administration of synthetic DNP decreased cardiac filling pressures; increased GFR, natriuresis, and diuresis; and lowered blood pressure, suppressing renin release and increasing plasma and urine cGMP levels.[291,292] Despite these propitious findings, several aspects of the biologic role of DNP remain contentious. In particular, the gene for the peptide has not been identified in mammals and the fractionation of DNP from human samples has not been reported.[282] These uncertainties have led some authors to question whether DNP is, in fact, expressed at all in humans.[282]

URODILATIN

Urodilatin is a structural homolog of ANP that shares the same 17–amino acid ring structure and COOH-terminal tail. It is synthesized in renal distal tubule cells and differentially processed to a 32–amino acid NH_2-terminally extended form of ANP.[293] Urodilatin is not found in plasma; instead, it acts in a paracrine manner within the kidneys on receptors in the glomeruli and IMCDs to promote natriuresis and diuresis. Urodilatin is upregulated in diabetic animals[294] and in the remnant kidney[295] and is relatively resistant to enzymatic degradation, which may explain its more potent renal effects.

NATRIURETIC PEPTIDE RECEPTORS

NPs mediate their biologic effects by binding to three distinct guanylyl cyclase NPRs. The terminology can be somewhat confusing: NPR-A binds ANP and BNP, and NPR-B binds CNP, whereas NPR-C acts as a clearance receptor for all three peptides.

NPR-A and NPR-B are structurally similar but share only 44% homology in the extracellular ligand-binding segment; this difference is probably responsible for the differences in ligand specificity. Both NPR-A and NPR-B have a molecular weight of approximately 120 kDa and consist of a ligand-binding extracellular domain, a single transmembrane segment, an intracellular kinase domain, and an enzymatically active guanylyl cyclase domain.[260] The kinase homology domain of NPR-A and NPR-B shares 30% homology with protein kinases but has no kinase activity. Ligand binding of NPR-A and NPR-B prevents the normal inhibitory action exerted by the kinase homology domain on the guanylyl cyclase domain, allowing the generation of cGMP, which acts as a second messenger responsible for most of the biologic effects of the NPs. NPR-C, in contrast to NPR-A and -B, lacks both the kinase homology domain and the catalytic guanylyl cyclase domain and therefore does not signal through a second-messenger system. Instead, the receptor contains the extracellular ligand-binding segment, a transmembrane domain, and a 37–amino acid cytoplasmic domain containing a G protein–activating sequence.[296] In NPR-C–knockout mice, blood pressure is reduced and the plasma half-life of ANP is increased; this finding supports the role of NPR-C as a clearance receptor.[297]

NPR-C binds all members of the NP family with high affinity. It is the most abundantly expressed of the NPRs—present in the kidneys, vascular endothelium, smooth muscle cells, and heart—and represents approximately 95% of the total receptor population. Preferential binding of NPR-C to ANP over BNP may explain the relatively increased plasma half-life of BNP.[270] NPR-C clears NPs from the circulation through a process of receptor-mediated endocytosis and lysosomal degradation before rapid recycling of the internalized receptor to the cell surface. Although the primary function of NPR-C is as a clearance receptor, ligand binding may exert biologic effects on the cell through G protein–mediated inhibition of cAMP.[298] The biologic effects of NPs are largely dependent on the distribution of their receptors. NPR-A mRNA is present mainly in the kidneys, especially in the IMCD cells, although the receptor is also notably present within the glomeruli, renal vasculature, and proximal tubules. The distribution of NPR-B overlaps with that of NPR-A; the receptor is found in the kidneys, vasculature, and brain. However, in accordance with the paracrine effects of CNP on vascular tone, mitogenesis, and cell migration, NPR-B is expressed in greater abundance than is NPR-A within the vascular endothelium and smooth muscle, whereas expression levels are relatively lower within the kidneys.

NEPRILYSIN

Receptor-mediated endocytosis probably accounts for about 50% of clearance of the NPs from the circulation; catalytic degradation by the enzyme neprilysin (NEP 24.11) is responsible for the majority of the rest, and direct renal excretion accounts for only a minor contribution.[270] Receptor clearance probably plays an even smaller role in conditions associated with chronically elevated NP levels, because of increased receptor occupancy and downregulation of NPR-C expression.

Neprilysin is a membrane-bound zinc metalloproteinase, originally termed *enkephalinase* because of its ability to degrade opioid receptors in the brain. The enzyme has structural and catalytic similarity to other metallopeptidases, including aminopeptidase; ACE; ECE; and carboxypeptidases A, B, and E and, in addition to the NPs, numerous other substrates have been described for neprilysin (Table 11.2). The primary mechanism of action of neprilysin is to hydrolyze peptide bonds on the NH_2 side of hydrophobic amino acid residues. In the case of ANP, neprilysin cleaves the Cys[105]-Phe[106] bond to disrupt the ring structure and inactivate the peptide. The Cys-Phe bond of BNP is relatively insensitive to enzymatic cleavage. Neprilysin has a nearly ubiquitous tissue distribution; expression has been demonstrated in the kidneys, liver, heart, brain, lungs, gut, and adrenal glands. The metallopeptidase is present not only on the surface of endothelial cells but also on smooth muscle cells, fibroblasts, and cardiac myocytes[299]; it is most abundant in the brush border of the proximal tubules of the kidneys, where it rapidly degrades filtered ANP, preventing the peptide from reaching more distal luminal receptors.

ACTIONS OF THE NATRIURETIC PEPTIDES

RENAL EFFECTS OF THE NATRIURETIC PEPTIDES

The natriuretic and diuretic actions of the NPs are consequences of both vasomotor effects and direct effects on the renal tubule. Both ANP and BNP cause an increase in glomerular capillary hydrostatic pressure and a rise in GFR by inducing afferent arteriolar vasodilation and efferent arteriolar vasoconstriction. These contrasting effects of the NPs on the afferent and efferent arterioles differ from the actions of classical vasodilators such as bradykinin. In addition to direct effects on vascular tone, ANP can increase GFR through cGMP-mediated mesangial cell relaxation and consequent changes in the ultrafiltration coefficient. Plasma levels of ANP that do not increase GFR can induce natriuresis, indicating the potential for direct tubule effects, which may involve either locally produced NPs acting in a paracrine manner, such as urodilatin, or circulating NPs. A number of mechanisms may be responsible for the natriuresis, including direct effects on sodium transport in tubule epithelial cells and indirect effects through inhibition of renin secretion after increased sodium delivery to the macula densa.

NPs also antagonize vasopressin in the cortical collecting ducts. Similar mechanisms probably underlie the response to ANP, BNP, and urodilatin. By contrast, CNP has little natriuretic or diuretic effect, which may indicate a requirement for the presence of the C-terminal extension of the peptide for renal effects. The NPs may have antifibrotic effects within the kidneys, as evidenced by an increase in renal fibrosis in NPR-A–knockout mice after unilateral ureteric obstruction.[300] In cultured proximal tubule cells, ANP attenuates high glucose–induced activation of TGF-β_1, Smad, and collagen synthesis, which illustrates the potentially antifibrotic properties of the peptide in the context of diabetic nephropathy.[301]

Table 11.2 Peptides That Have Been Described as Substrates for Neprilysin[393,547]

Atrial natriuretic peptide	Cholecystokinin	Interleukin-1β
Brain natriuretic peptide	Corticotropin-releasing hormone	β-Lipotropin
C-type natriuretic peptide	Dynorphins	Luliberin
Endothelin-1	Endorphins	Luteinizing hormone-releasing hormone
Bradykinin and kallidin	Endothelin-2	α-Melanocyte-stimulating hormone
Substance P	Endothelin-3	Neurokinin A
Angiotensin I	Enkephalins	Neuropeptide Y
Angiotensin II	N-Formylmethionine-leucyl-phenylalanine	Neurotensin
Angiotensin 1-7	Fibroblast growth factor-2	Oxytocin
Adrenocorticotrophic hormone	Gastric-inhibitory peptide	Peptide YY
Adrenomedullin	Gastrin-releasing peptide	Secretin
Amyloid-β peptide	Glucagon	Somatostatin
Big endothelin-1	Gonadotropin-releasing hormone	Thymopentin
Bombesin-like peptides	Incretins	Vasoactive intestinal peptide
Calcitonin gene–related peptide	Insulin B chain	Vasopressin

CARDIOVASCULAR EFFECTS OF THE NATRIURETIC PEPTIDES

All NPs have vasodilatory and hypotensive properties. Heterozygous mutant mice with a disrupted proANP gene display evidence of salt-sensitive hypertension,[302] whereas hypotension is a feature of transgenic mice overexpressing ANP.[303] In a human patient population, a variant in the ANP promoter was associated with both lower levels of plasma ANP and increased susceptibility to early development of hypertension.[304] However, infusion of high concentrations of ANP can actually induce a rise in blood pressure, which suggests that counterregulatory baroreceptors may be activated.[305]

ANP lowers blood pressure through two major direct mechanisms. First, it increases vascular permeability with a shift of fluid from the intravascular to extravascular compartments by capillary hydraulic pressure. Second, ANP increases venous capacitance and lowers preload.[306] In addition, ANP and BNP antagonize the vasoconstrictive effects of the RAS, ET, and the sympathetic nervous system[265] by decreasing sympathetic peripheral vascular tone, thereby suppressing the release of catecholamines and reducing central sympathetic outflow.[267] By lowering the activation threshold of vagal afferents, ANP prevents the vasoconstriction and tachycardia that normally follow a reduction in preload and thereby produces a sustained drop in blood pressure. CNP is a more potent vasodilator than either ANP or BNP. In fact, CNP relaxes human subcutaneous resistance arteries, whereas ANP and BNP have no effect.[307]

NPs have a number of other effects on the cardiovascular system distinct from their action on vasomotor tone. For example, NPs play a major role in cardiac remodeling. Mice with genetic deficiencies of ANP exhibit an increase in cardiac mass,[302] whereas heart size is diminished in mice transgenically overexpressing ANP.[303] The antimitogenic and antitrophic effects of NPs, which appear to be mediated by cGMP, have also been demonstrated in a range of cultured cell types, including cultured vascular cells, fibroblasts, and myocytes, and in vivo in response to balloon angioplasty. Further evidence for the role of ANP in mediating cardiac hypertrophy was obtained from population studies, in which variants in either the *NPPA* promoter (associated with reduced circulating ANP) or the NPR-A gene, NPR1, have been associated with left ventricular hypertrophy.[308,309] BNP has been shown to have antifibrotic properties within the heart. In vitro, BNP antagonizes TGF-β–induced fibrosis in cardiac fibroblasts,[310] and in vivo, targeted genetic disruption of BNP in mice is associated with an increase in cardiac fibrosis, in the absence of either hypertension or ventricular hypertrophy.[311]

Cardiac CNP is increased in heart failure where it may play a role in ventricular remodeling.[312] Comparison of plasma CNP levels in samples taken from the aorta and renal vein, at the time of diagnostic heart catheterization, has demonstrated that CNP is indeed synthesized and secreted by the kidney.[313] Moreover, this effect was found to be blunted in patients with heart failure, potentially contributing to renal sodium retention.[313] In rats subjected to unilateral ureteric obstruction, recombinant CNP decreased blood urea nitrogen and creatinine levels and attenuated renal fibrosis.[314]

OTHER EFFECTS OF THE NATRIURETIC PEPTIDES

Even though they do not cross the blood–brain barrier, NPs exert important CNS effects that may augment their peripheral actions. ANP, BNP, and particularly CNP are all expressed within the brain. Circulating NPs may also exert central effects through actions at sites that are outside the blood–brain barrier. The NPR-B receptor is expressed throughout the CNS, which reflects the wide distribution of CNP, whereas the NPR-A receptor is expressed in areas adjacent to the third ventricle, which is indicative of a role of peripherally circulating ANP and BNP, as well as centrally expressed peptides. Complementing their natriuretic and diuretic effects, NPs inhibit both salt appetite and water drinking. ANP also prevents release of vasopressin and possibly adrenocorticotropic hormone from the pituitary gland, whereas sympathetic tone is increased by the actions of the NPs on the brain stem.

Clinical and experimental evidence suggests that NPs play a role in mediating metabolism. Circulating levels of NPs are decreased in obese individuals[315] and among patients with the metabolic syndrome,[316,317] correlating inversely with both plasma glucose and fasting insulin levels.[318] In accordance with these epidemiologic observations, infusion of ANP activates hormone-sensitive lipase from fat cells, which is indicative of lipolysis.[319] In vitro, ANP inhibits preadipocyte proliferation,[320] the lipolytic properties of the peptide being mediated by cGMP phosphorylation.[321,322]

Knockout mouse studies have revealed that CNP plays a predominant role in the regulation of skeletal growth, specifically cartilage homeostasis and endochondral bone formation.[323] Mice with genetic deficiencies of either CNP or its receptor NPR-B lack growth of longitudinal bones and vertebrae and have a shortened life span as a consequence of respiratory insufficiency secondary to abnormal ossification of the skull and vertebrae.[324,325] Transgenic mice that overexpress CNP are relatively protected from glucocorticoid-induced growth retardation.[326] Mutations in the NPR-B gene have also been reported in patients with the autosomal recessive skeletal dysplasia and acromesomelic dysplasia–type Maroteaux, and obligate carriers of the mutations have heights that are below predicted levels.[325] Accordingly, CNP analog therapy is being investigated as a possible treatment for achondroplasia.[327]

NATRIURETIC PEPTIDES AS BIOMARKERS OF DISEASE

Both ANP and BNP have been studied as clinical biomarkers of heart failure and renal failure. The short half-life of ANP (2–5 minutes) restricts its applicability.[328] However, the biologically inactive NH_2-terminal 98–amino acid peptide ANP 1-98 does not bind to NPR-A or NPR-C and so remains in the circulation longer than ANP does. In heart failure, ANP 1-98 levels closely reflect the degree of renal function.[329] Plasma concentrations of the midregional epitopes of the stable prohormones of both ANP and adrenomedullin are predictive of the progression of renal decline in patients with nondiabetic CKD.[330] The prognostic performance of midregional proANP is not superior to that of NT-proBNP or BNP in hemodialysis patients[331] and measurement of ANP or one of its prohormone derivatives is currently not part of routine clinical care. Signal peptides from both ANP and BNP are present in venous blood and rise rapidly following myocardial infarction, suggesting that their detection may aid in the diagnosis of cardiac ischemia.[332,333] Commercial assays are widely available for measurement of either BNP or the biologically inactive peptide fragment NT-proBNP. Correspondingly, since 2000, measurement of circulating BNP and NT-proBNP levels has been incorporated into several clinical practice guidelines for the management of heart failure. Important differences distinguish BNP and NT-proBNP from each other as clinical biomarkers. NT-proBNP is not removed from the circulation by binding to the clearance receptor NPR-C, and hence its circulating half-life of approximately 2 hours is significantly longer than that of BNP (approximately 20 minutes). In addition, both BNP and NT-proBNP are affected by renal impairment,[334] but the magnitude of the effect is greater for NT-proBNP.[335]

BRAIN NATRIURETIC PEPTIDE AND N-TERMINAL PROBRAIN NATRIURETIC PEPTIDE AS BIOMARKERS OF HEART FAILURE

Measurement of circulating levels of either BNP or NT-proBNP has effectively helped guide clinical practice in several aspects of the management of heart failure, including diagnosis, screening, prognosis, and monitoring of therapy.[260] The primary role of BNP measurement in the assessment of dyspnea is as a “ruling out” test: A plasma BNP level lower than 100 pg/mL has a negative predictive value for heart failure of 90%.[336] In the ProBNP Investigation of Dyspnea in the Emergency Department (PRIDE) study, an NT-proBNP level lower than 300 pg/mL was optimal in ruling out heart failure, with a negative predictive value of 99%.[337] Screening for BNP[338] and NT-BNP[339] levels has also proven useful in identifying individuals at risk for heart failure for whom aggressive medical therapy should be targeted. The utility of serial BNP or NT-BNP measurements in guiding the treatment of patients with heart failure was the subject of a 2016 Cochrane Review.[340] This review concluded that low-quality evidence shows that NP-guided treatment is associated with a reduction in hospital admissions for heart failure and that low-quality evidence shows uncertainty with respect to the effect of NP-guided treatment on mortality or all-cause hospital admissions.[340] In the interpretation of plasma levels of BNP and NT-proBNP, a number of other biologic variables should be taken into account. NP levels rise with age and are higher in women, the latter effect possibly secondary to estrogen regulation, inasmuch as hormone replacement therapy increases BNP levels.[341] Conversely, NP levels fall with increasing obesity. Although BNP levels of heart failure patients are higher in Asian or African American patients than in Caucasian or Hispanic patients, they provide prognostic value regardless of race or ethnicity.[342]

ROLE OF BRAIN NATRIURETIC PEPTIDE AND N-TERMINAL PROBRAIN NATRIURETIC PEPTIDE AS BIOMARKERS IN RENAL DISEASE

The interpretation of NP concentrations in patients with renal disease merits special consideration. NP levels are increased in individuals with impaired renal function. This increase is probably multifactorial in origin and not solely the consequence of increased intravascular volume. Other factors that contribute to increased NP levels include decreased NP responsiveness, subclinical ventricular dysfunction, hypertension, left ventricular hypertrophy, subclinical ischemia, myocardial fibrosis, and RAS activation,[343] as well as decreased filtration and reduced clearance by NPR-C and NEP.[344] Although, on the basis of observational studies, it has been widely considered that renal clearance plays a greater role in the removal of NT-proBNP from the circulation than removal of BNP, one study has challenged this view. By measuring both NT-proBNP and BNP in the renal arteries and veins of 165 subjects undergoing renal arteriography, investigators found that both NT-proBNP and BNP are equally dependent on renal clearance.[345] However, the NT-proBNP-to-BNP ratio did increase with declining GFR, which suggests that the two peptides may be differentially cleared at GFRs lower than 30 mL/min/1.73 m^2.[345]

Even though both BNP and NT-proBNP are affected by renal impairment, their clinical utility for the prediction of heart failure persists in CKD patients in the context of appropriately adjusted reference ranges. For example in the Breathing Not Properly study, BNP cut point values were approximately threefold higher to diagnose heart failure in patients with an estimated GFR lower than 60 mL/min relative to the conventional cut point value of 100 pg/mL.[346] In a cohort of 831 patients with dyspnea and a GFR less than 60 mL/min, both BNP and NT-proBNP were effective predictors of heart failure, although NT-proBNP was superior in predicting mortality.[347] In asymptomatic patients with CKD, both BNP and NT-proBNP were equivalent and effective in indicating the presence of left ventricular hypertrophy or coronary artery disease.[348] In patients with CKD, BNP and NT-proBNP may be predictive of the progression of renal decline[349,350] and cardiovascular disease and mortality. In a nondialysis CKD population, NT-proBNP, but not BNP, was an independent predictor of death[351]; in 994 black patients with hypertensive renal disease (GFR = 20 to 65 mL/min/1.73 m^2), NT-proBNP was predictive of cardiovascular disease and mortality, particularly among individuals with proteinuria.[352] In pediatric CKD patients, both BNP and pro-BNP (but not troponins I and T) were indicative of left ventricular hypertrophy or dysfunction.[353]

BNP and NT-proBNP have been studied extensively in dialysis recipients both as prognostic indicators and as markers of volume status. The molecular weights of BNP (3.5 kDa) and NT-proBNP (8.35 kDa) are low enough that both peptides may be cleared by high-flux dialysis.[354,355] Nevertheless, in contrast to ANP, which falls sharply after either hemodialysis or peritoneal dialysis, levels of BNP and NT-proBNP are less affected.[354,356] The role of NP levels as indicators of volume status in either hemodialysis or peritoneal dialysis recipients is confounded by the common coexistence of left ventricular abnormalities.[334,346,357–360] Both BNP and NT-proBNP levels are predictive of mortality, heart failure, and coronary artery disease in the population undergoing dialysis.[361–366] However, no definite cut point values for diagnosing heart failure in dialysis patients have been defined.[346]

CIRCULATING C-TYPE NATRIURETIC PEPTIDE LEVELS AS A BIOMARKER FOR RISK OF MYOCARDIAL INFARCTION

Although CNP usually functions in a paracrine manner, its presence in the plasma may provide utility as a biomarker of cardiovascular risk. In a study of 1841 individuals from the general population, individuals with plasma CNP levels in the highest quartile were at increased risk of myocardial infarction and, unlike BNP levels, plasma CNP levels were unaffected by sex and only weakly associated with age.[367]

THERAPEUTIC USES OF NATRIURETIC PEPTIDES

Even though NP levels are increased in heart failure, their biologic effects are blunted. Intravenous administration of recombinant NPs increases their circulating levels several-fold, overcoming this resistance. As such, two recombinant NPs are currently available as therapeutic agents for the treatment of heart failure: recombinant ANP (carperitide), which is available in Japan for the treatment of pulmonary edema, and recombinant BNP (nesiritide), which is licensed in several countries, including the United States, for the treatment of acute decompensated heart failure.

RECOMBINANT ATRIAL NATRIURETIC PEPTIDE

ANP has a short half-life and a high total body clearance. Its intravenous administration causes a reduction in blood pressure, diuresis, and natriuresis in healthy individuals; this response is reduced in the setting of acute heart failure. In a 6-year open-label study of 3777 patients with acute heart failure treated with carperitide, clinical improvement was reported in 82%.[368] Whereas early experimental studies were suggestive of a potential benefit of exogenous ANP in acute renal failure, results in patients have generally been disappointing. Nevertheless, the peptide may have a limited role in selected patient populations. For example, low-dose carperitide preserved renal function in patients undergoing repair of abdominal aortic aneurysm[369] and reduced the incidence of contrast-induced nephropathy in patients after coronary angiography.[370] However, a meta-analysis suggested that recombinant ANP has no effect on mortality in patients with acute renal injury, although a trend toward a reduction in the need for renal replacement therapy was shown.[371] In a separate meta-analysis of studies conducted in cardiovascular surgery patients, ANP infusion decreased peak serum creatinine, incidence of arrhythmia, and need for renal replacement therapy, whereas both ANP and BNP decreased the length of intensive care unit and hospital stay.[372] Among 367 high-risk individuals undergoing coronary artery bypass grafting (CABG), recombinant ANP decreased the incidence of major adverse cardiovascular and cerebrovascular events and the need for dialysis, immediately and up to 2 years postoperatively, although survival was unaffected.[373] Similarly, among CKD patients undergoing CABG, those receiving recombinant ANP experienced a smaller rise in serum creatinine, fewer cardiac events, and lower requirement for dialysis, although mortality did not differ from those that did not receive ANP.[374] However, when employed in an effort to treat rather than prevent acute kidney injury following cardiac surgery, recombinant ANP had no significant effect on renal function, the need for renal replacement, length of stay, or medical costs.[375]

RECOMBINANT BRAIN NATRIURETIC PEPTIDE

Nesiritide is recombinant human BNP, manufactured from *Escherichia coli* and identical in structure to native human BNP, with a mean terminal half-life of 18 minutes in patients with heart failure.[376] Intravenous administration of nesiritide lowers pulmonary and systemic vascular resistance, decreases right atrial pressure, and increases cardiac output (presumably through effects on ventricular afterload) in a concentration-dependent fashion.[377] In the kidneys, nesiritide increases RBF and GFR through both direct vasodilatory effects and indirect effects on cardiac output and norepinephrine inhibition.[378] Diuresis and natriuresis may also occur, although these effects are modest and may not be seen at the approved doses. Additional effects of nesiritide may also include inhibition of renin secretion in the kidneys and aldosterone production in the heart and adrenal glands.

In response to meta-analysis data suggesting that nesiritide treatment may be associated with a worsening of renal function and an increase in the rate of early death,[379,380] the Acute Study of Clinical Effectiveness of Nesiritide in Decompensated Heart Failure (ASCEND-HF) trial was initiated.[381] In this study of 7141 patients hospitalized with acute heart failure, nesiritide neither increased nor decreased the rate of death or rehospitalization, rates of worsening renal function were unaffected, and there was a small but nonsignificant improvement in self-reported rates of dyspnea.[381] Based on these results, the investigators concluded that nesiritide cannot be recommended for routine use in the broad population of patients with acute heart failure.[381]

THERAPEUTIC USES OF OTHER NATRIURETIC PEPTIDES

The effects of urodilatin (ularitide) have been assessed in both heart failure and acute renal failure. However, the diuretic effect of urodilatin appears to be attenuated in heart failure patients, which reflects a blunted response, as observed for ANP and BNP.[382,383] Similarly, as with ANP and BNP, hypotension appears to be a dose-limiting side effect of ularitide therapy.[383,384] In the Safety and Efficacy of an Intravenous Placebo-Controlled Randomized Infusion of Ularitide in a Prospective Double-blind Study in Patients with Symptomatic, Decompensated Chronic Heart Failure (SIRIUS II) study, a phase II trial of 221 patients hospitalized for decompensated heart failure, a single 24-hour infusion of ularitide preserved short-term renal function.[385] The NP vessel dilator may offer theoretical advantages for the treatment of acute decompensated heart failure in comparison with current NP-based therapies.[386,387] In particular, vessel dilator may produce a greater and more sustained natriuresis than does ANP or BNP, without a blunted response in patients with heart failure, and may also improve renal function in the setting of experimental acute renal injury.[388] An alternative therapeutic approach is the development of novel chimeric peptides. For example, researchers have synthesized a peptide (cenderitide) that represents fusion of the 22–amino acid peptide CNP together with the 15–amino acid linear C terminus of DNP.[389] In vitro, this peptide activates cGMP and attenuates cardiac fibroblast proliferation. In vivo, cenderitide is both natriuretic and diuretic and increases GFR with less hypotension than does BNP.[389,390] Cenderitide is more resistant to degradation by neprilysin than the naturally occurring NPs and is eightfold more potent in inducing glomerular cGMP production than CNP.[391]

COMBINATION ANGIOTENSIN RECEPTOR BLOCKADE AND NEPRILYSIN INHIBITION

Notwithstanding concerns regarding the efficacy and cost effectiveness of recombinant NP therapy, a major limitation is the requirement for systemic administration, which is unsuitable for chronic treatment. Alternative methods to increase the biologic activity of NPs may offer a more feasible approach for chronic therapy. In particular, inhibition of the enzymatic degradation of NPs by neprilysin has been the focus of drug discovery efforts for a number of years. Neprilysin is a zinc metallopeptidase with catalytic similarity to ACE and with a wide tissue distribution, although abundant at the proximal tubule brush border. Several pharmacologic neprilysin inhibitors have been investigated (e.g., candoxatril, thiorphan, and phosphoramidon). Although these agents, in general, lead to an increase in plasma levels of the NPs and, under some experimental conditions, induce natriuresis and diuresis with peripheral vasodilation, results of clinical trials in hypertension and heart failure have generally been disappointing. Specifically, sustained antihypertensive effects have not been demonstrated, and some researchers have reported a paradoxical rise in blood pressure. This may be a consequence of the induction of both neprilysin and ACE expression with neprilysin inhibition[392] and a consequent increase in angiotensin II levels.[393] The biologic actions of the NPs are, however, restored in the presence of an inhibited RAS and this has led to the development of two classes of agent: (1) vasopeptidase inhibitors that inhibit both neprilysin and ACE and (2) combined angiotensin receptor blockade/neprilysin inhibition, the latter having gained regulatory authority approval for the treatment of heart failure.

The rational design of vasopeptidase inhibitors—such as mixanpril (S21402), CGS30440, aladotril, MDL 100173, sampatrilat, and omapatrilat—was made possible because of the similar structural characteristics of the catalytic sites of both neprilysin and ACE.[299] Despite the theoretical advantages of vasopeptidase inhibitors, phase III clinical studies have not been able to demonstrate superiority of vasopeptidase inhibition over ACE inhibition, and an increase in the incidence of angioedema has raised safety concerns. For instance, in the Omapatrilat Cardiovascular Treatment vs. Enalapril (OCTAVE) trial of 25,302 hypertensive patients, angioedema occurred in 2.17% of omapatrilat-treated patients, in comparison with 0.68% of patients treated with the ACE inhibitor enalapril.[394] In the Omapatrilat Versus Enalapril Randomized Trial of Utility in Reducing Events (OVERTURE) study, the incidence of angioedema was, again, increased among subjects receiving omapatrilat in comparison with those receiving enalapril (0.8% vs. 0.5%).[395] The increased angioedema with vasopeptidase inhibition is likely a consequence of decreased degradation of bradykinin and substance P with combined inhibition of the two metallopeptidases.[396]

The rationale that concurrent RAS blockade may potentiate the therapeutic effects of neprilysin inhibition yet concurrent ACE inhibition increases the risk of angioedema encouraged the development of a new class of drug that combines an ARB and neprilysin inhibitor. This class has been termed ARNi (i.e., angiotensin receptor-neprilysin inhibitor), although ARBs do not inhibit enzymatic activity. In July 2015, the first in the class, valsartan/sacubitril gained approval from the U.S. Food and Drug Administration for the treatment of heart failure with reduced ejection fraction. Valsartan/sacubitril is a single molecule composed of molecular moieties of the ARB, valsartan, and the neprilysin inhibitor prodrug sacubitril (formerly AHU-377) in a 1:1 ratio.[397] During development, the combination drug of valsartan/sacubitril was referred to as LCZ696. After ingestion, valsartan/sacubitril dissociates into valsartan and sacubitril, and sacubitril is subsequently converted to its active form sacubitrilat (LBQ657) by esterases.[396] In a study of 1328 patients, valsartan/sacubitril conferred greater blood pressure lowering than valsartan alone with no cases of angioedema reported.[398] In the Prospective comparison of ARNi with ARB on Management Of heart failUre with preserved ejectioN fracTion (PARAMOUNT) study of 301 individuals with heart failure with preserved ejection fraction, valsartan/sacubitril lowered NT-proBNP levels to a greater extent than valsartan after 12 weeks of treatment and was well tolerated.[399] Although the reduction in NT-proBNP was sustained at 36 weeks, the difference in NT-proBNP levels between participants randomized to valsartan/sacubitril and valsartan was no longer significant.[399] However, left atrial remodeling and heart failure symptoms were improved.[399]

The case for regulatory authority approval for valsartan/sacubitril was based on the findings of the phase III prospective comparison of ARNi with ACEi (Determine Impact on Global Mortality and Morbidity in Heart Failure [PARADIGM-HF] trial).[400] PARADIGM-HF compared the effects of valsartan/sacubitril (200 mg twice daily) and enalapril (10 mg twice daily) in 8442 patients with heart failure (New York Heart Association class II–IV) and a reduced ejection fraction (≤40%).[400] The primary outcome, a composite of cardiovascular death and hospitalization for heart failure, occurred in 21.8% of participants treated with valsartan/sacubitril and 26.5% of participants treated with enalapril (hazard ratio 0.80, confidence interval 0.73–0.87, $P < .001$). Hypotension was more common in participants receiving valsartan/sacubitril, whereas cough, hyperkalemia, and renal impairment were more common in those receiving enalapril.[400] Importantly, there was no significant difference in the number of cases of angioedema between participants receiving valsartan/sacubitril than those receiving enalapril, although the number of cases of angioedema was numerically higher in the valsartan/sacubitril group (19 vs. 10, $P = .13$). The effect of valsartan/sacubitril in patients with heart failure and preserved ejection fraction is currently being evaluated in the Prospective Comparison of ARNI with ARB Global Outcomes in Heart Failure with preserved Ejection Fraction (PARAGON-HF) trial (NCT01920711), and other molecules combining ARB and NEP inhibition are also currently under development.[401,402]

Despite the promising findings of PARADIGM-HF, there are some difficulties with the design of the study and there are some theoretical considerations around the use of neprilysin inhibitors in a broad population. In terms of the active comparator in PARADIGM-HF, it is noteworthy that valsartan/sacubitril was not compared with valsartan alone and that the dose of enalapril (10 mg twice daily) may have been insufficient. For instance, in the Cooperative North Scandinavian Enalapril Survival Study (CONSENSUS) of patients with heart failure, the target dose of enalapril was up to 20 mg twice daily.[403] Separately, whether the apparently low risk of angioedema observed in PARADIGM-HF translates to the real-world setting remains to be determined. Only 5% of participants in PARADIGM-HF were black, a group at increased risk of angioedema associated with ACE inhibition or neprilysin inhibition.[396] Furthermore, 78% of participants in PARADIGM-HF had been previously treated with an ACE inhibitor and inclusion of a run-in period where all participants were exposed to enalapril may have resulted in an underrepresentation in the number of cases of angioedema.[400]

Other theoretical risk concerns surrounding chronic neprilysin inhibition largely relate to the other substrates that are normally degraded by neprilysin (Table 11.2). Based on the breadth of activity of these substrates, the possibility has been raised that long-term neprilysin inhibition could have deleterious effects on bronchial reactivity, pain, inflammation, tumorigenesis, and neuronal function.[393] Of particular note has been the recognition that neprilysin is important in the metabolism of amyloid-β peptides and that its inhibition could predispose to the development of Alzheimer's disease, age-related macular degeneration, and cerebral amyloid angiopathy which may take many years to manifest.[393] A trial comparing the efficacy and safety of valsartan/sacubitril with valsartan on cognitive function in patients with heart failure and preserved ejection fraction is ongoing (NCT02884206).

RENAL EFFECTS OF VALSARTAN/SACUBITRIL IN PATIENTS WITH HEART FAILURE AND PRESERVED EJECTION FRACTION

The renal effects of valsartan/sacubitril were assessed in a post hoc analysis of participants in the PARAMOUNT trial of individuals with heart failure and preserved ejection fraction.[404] Participants received treatment with valsartan/sacubitril titrated to 200 mg twice daily or valsartan titrated to 160 mg twice daily, which each give similar levels of systemic exposure to valsartan.[397,398] In the PARAMOUNT trial, eGFR declined less with valsartan/sacubitril treatment than it did with valsartan treatment over a 36-week period (–1.5 vs. –5.2 mL/min/1.73 m^2, $P = .002$), whereas the geometric mean of the urinary albumin-to-creatinine ratio increased from baseline in the valsartan/sacubitril group (2.4–2.9 mg/mmol) and was unchanged in the valsartan group (2.1–2.0 mg/mmol; P value for difference between groups = .016).[404] The finding of a relative preservation in eGFR with valsartan/sacubitril is consistent with the observation from the PARADIGM-HF trial that valsartan/sacubitril-treated patients experienced less renal impairment that necessitated cessation of therapy than enalapril-treated patients.[400] Both the effect on eGFR and albuminuria are reminiscent of the effect of systemic ANP administration, raising the possibility that they are a consequence of increased levels of biologically active ANP with neprilysin inhibition.[404] Participants in the PARAMOUNT trial were required to have an eGFR of at least 30 mL/min/1.73 m^2 at enrollment[399] and thus the effects of combination ARB and neprilysin inhibition in more advanced renal disease are currently unknown, as are the effects on hard renal endpoints in at-risk populations.

OTHER NATRIURETIC PEPTIDES

GUANYLIN AND UROGUANYLIN

The existence of intestinal NPs has been suggested by initial observations that sodium excretion is greater after an oral salt load than after an intravenous salt load.[405,406] These intestinal peptides include guanylin and uroguanylin. However, a study of 15 healthy volunteers found that sodium excretion was similar in response to either oral or intravenous sodium load during either a low- or high-sodium–containing diet.[407] Moreover, serum concentrations of either prouroguanylin or proguanylin were unchanged following either oral or intravenous sodium load and showed no correlation with sodium excretion.[407] Collectively, these observations challenge the notion of a gastrointestinal–renal natriuretic axis mediated by the guanylin peptide family.[407,408] It thus appears likely that the natriuretic, kaliuretic, and diuretic effects of guanylin and uroguanylin, which occur without change in GFR or RBF, are mediated by local production of the peptides within the kidney.[408]

ADRENOMEDULLIN

Adrenomedullin is a 52–amino acid peptide originally isolated from human pheochromocytoma cells,[409] although it is synthesized mainly by vascular smooth muscle cells, endothelial cells, and macrophages[410] and is present in the plasma, vasculature, lungs, heart, and adipose tissue. The peptide is upregulated in patients with cardiovascular disease and has positive inotropic and vasodilatory properties. Systemic administration of adrenomedullin induces an NO-dependent natriuresis and an increase in GFR both under normal conditions and in patients with congestive heart failure; it also decreases plasma aldosterone levels without affecting renin activity. Individuals with type 2 diabetes and plasma levels of midregional proadrenomedullin (MR-proADM) peptide in the highest tertile are at an increased risk of severe nephropathy (doubling of plasma creatinine and/or ESRD), which may reflect a reactive rise in MR-proADM.[411]

KALLIKREIN–KININ SYSTEM

The KKS is a complex network of peptide hormones, receptors, and peptidases that is evolutionarily conserved with homologs in nonmammalian species.[412] Discovery of the KKS is attributed to Abelous and Bardier, who reported in 1909 that experimental injection of urine resulted in an acute fall in systemic blood pressure.[412a] Since that time, investigators have recognized that the physiologic actions of the KKS also include regulation of tissue blood flow, transepithelial water and electrolyte transport, cellular growth, capillary permeability, and inflammatory responses. The main components of the KKS are the enzyme kallikrein, its substrate kininogen, effector hormones known as *kinins* (especially bradykinin and kallidin [also termed *lys-bradykinin*]) and their inactivating enzymes, which include kininases I and II (ACE) and neprilysin.

Kinins exert their biologic effects through binding to two receptors: the bradykinin B1 receptor (B1R) and bradykinin B2 receptor (B2R). The B2R is widely expressed and mediates all the physiologic actions of the kinins under normal conditions. The B1R is activated predominantly by des-Arg-bradykinin, a natural degradation product of bradykinin, generated by cleavage of the peptide by kininase I. The KKS may be subdivided into a circulatory (plasma) KKS and a tissue (including renal) KKS, which may be distinguished by their principal effector molecules, bradykinin and kallidin, respectively. In the kidneys, the kinins play a significant role in the modulation of renal hemodynamics and salt and water homeostasis.

COMPONENTS OF THE KALLIKREIN–KININ SYSTEM

KININOGEN

Humans possess a single kininogen gene, *KNG1,* which is localized to chromosome 3q26 and encodes both high–molecular weight (HMW) kininogens (626 amino acids, 88 to 120 kDa) and low–molecular weight (LMW) kininogens (409 amino acids, 50–68 kDa) through alternate splicing from 11 exons spread over a 27-kb genomic region. A second kininogen gene has been identified in mice.[413] In humans, kininogen deficiency may be relatively asymptomatic[414]; the kininogen-deficient Brown Norway Katholiek rat strain, however, shows increased sensitivity to the pressor effects of salt, angiotensin II, and mineralocorticoid.[415,416]

KALLIKREIN

HMW and LMW kininogen are cleaved by the serine protease kallikrein. The name "kallikrein" is derived from the Greek term *kallikreas,* meaning "pancreas," after the work of Frey and others, in the 1930s, who extracted a kinin-producing enzyme from the pancreas of dogs.[416a] Since then, 15 tissue kallikreins have been identified, although, in humans, only one (KLK1) is involved in local kinin production. The human kallikrein genes are clustered on chromosome 19 at loci q13.3-13.4. Plasma kallikrein is found in the circulation and is involved largely with the coagulation cascade and activation of neutrophils. The tissue kallikreins are acid glycoproteins that are variably and extensively glycosylated. Human renal kallikrein is synthesized as a zymogen (prekallikrein) with a 17–amino acid signal peptide and a 7–amino acid activation sequence, which must be cleaved in order to activate the enzyme. In most mammals, including humans, tissue kallikrein cleaves kallidin (lys-bradykinin) from kininogens, whereas plasma kallikrein releases bradykinin.

Although the physiologic effects of kallikrein have been attributed to increased kinin generation, the enzyme may also have direct effects on the B2R, as well as actions independent of the kinin receptors.[417,418] For example, in kininogen-deficient Brown Norway Katholiek rats, local injection of kallikrein into the myocardium after coronary artery ligation had a cardioprotective effect that was abolished by the NO synthase inhibitor Nω-nitro-L-arginine methyl ester and the selective B2R inhibitor icatibant (Hoe 140).[417] As a serine protease, kallikrein may also elicit kinin receptor–independent effects on endothelial cell migration and survival through cleavage of growth factors and matrix metalloproteinases.[419] Transgenic mice overexpressing human kallikrein exhibit a sustained reduction in systemic blood pressure throughout their life span, which is indicative of the lack of sufficient compensatory mechanisms to reverse the hypotensive effect of kallikrein.[420] In humans, polymorphisms of the kallikrein gene *KLK1* or its promoter can impair enzymatic activity, potentially influencing both kinin-dependent and kinin-independent effects. Among normotensive men with a common loss-of-function *KLK1* polymorphism (R53H), an increase in wall shear stress and a paradoxical reduction in artery diameter and lumen were noted, although flow-mediated and endothelium-independent vasodilation were unaffected.[421]

KININS

The kinins are bradykinin and kallidin in humans and bradykinin and kallidin-like peptide in rodents.[422] Plasma aminopeptidase can convert kallidin (10 amino acids: Lys-Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg) to bradykinin (9 amino acids: Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg) by cleavage of the first N-terminal lysine residue. Cleavage of the carboxy-terminal arginine residue by kininase I (carboxypeptidase-N) and carboxypeptidase-M generates their des-Arg derivatives, which are agonists of the B1R.[422] Removal of two C-terminal amino acids (Phe and Arg) by ACE (kininase II), neprilysin, or ECE is responsible for inactivation of the peptides.[422]

BRADYKININ RECEPTORS

B1R and B2R share 36% homology, and both are GPCRs with seven transmembrane domains. The genes for the two receptors are in tandem on a compact locus (14q23) separated by only 12 kb.[423] The B2R is the principal receptor mediating the actions of both kinins, is expressed in abundance by vascular endothelial cells, and is present in most tissues, including those of the kidneys, heart, skeletal muscle, CNS, vas deferens, trachea, intestines, uterus, and bladder. In general, the distribution and action of B1Rs are similar to those of the B2Rs. The B1R, by contrast, is expressed at low levels under normal conditions but is upregulated in response to inflammatory stimuli (e.g., lipopolysaccharide, endotoxins, and cytokines such as IL-1β and TNF-α)[424] and in the setting of diabetes[425] and ischemia-reperfusion injury.[426] B2R binds both bradykinin and kallidin, whereas bradykinin has almost no effect at the B1R. The carboxypeptidase required to generate the des-Arg B1R-active kinin fragments is closely associated with the B1R on the cell surface.[427] This association would enable B2R agonists to rapidly activate B1Rs, particularly in response to inflammation.[427]

Ligand binding of both receptor subtypes induces activation of phospholipase C, which results in intracellular calcium mobilization through production of inositol 1,4,5-triphosphate and DAG via activation of G proteins, including $G\alpha_q$ and $G\alpha_i$. The physiologic effects of bradykinin receptor activation are mediated through generation of both endothelial NO synthase–derived NO and prostaglandins. B2R activation leads to a rise in intracellular calcium concentrations in vascular endothelial cells.[422] However, bradykinin-induced vasodilation is not abolished by coadministration of NO synthase and COX inhibitors, which indicates that additional effectors are also likely to be involved, possibly an endothelium-derived hyperpolarizing factor. In addition, through binding to both B1R[428] and B2R,[429] bradykinin also increases the expression of inducible NO synthase (iNOS), at least in rodents. It is very difficult to induce the iNOS gene in human tissues, especially the vascular endothelium. Mice that have genetic deficiencies of B2R,[430] B1R,[431] or both receptors[432] have been generated; the reported phenotypes of the different knockout strains have been varied, which may be a result of different genetic backgrounds, or, in the case of the single knockouts, differing compensatory effects of the remaining receptor. For example, some studies of B2R-deficient mice revealed an increase in resting systemic blood pressure, an exaggerated pressor response to angiotensin II[433] and salt sensitivity,[434] whereas others revealed no difference in resting blood pressure between B2R- or B1R-deficient mice and wild-type animals.[431,435] Double B2R-/B1R-knockout mice were also reported to have resting blood pressure identical to that in wild-type mice and were resistant to lipopolysaccharide-induced hypotension.[432,436] By contrast, transgenic mice expressing the human B2R had a lower resting blood pressure than did wild-type controls.[437] Transgenic mice expressing the rat B1R (as well as their native murine B2R) were normotensive but showed an exaggerated hypotensive response to lipopolysaccharide and, unexpectedly, a hypertensive response to des-Arg bradykinin.[438]

KALLISTATIN

Kallistatin is an endogenous serpin inhibitor of kallikrein that acts by forming a heat-stable complex with the enzyme. Surprisingly, administration of human kallistatin to rodents induced vasodilation and a decline in systemic blood pressure, which was unaltered by either an NO synthase inhibitor or the B2R antagonist icatibant; this suggests that the vasodilatory properties of kallistatin may be mediated through a smooth muscle mechanism independent of bradykinin receptor activation.[439]

KININASES

With the exception of the metabolites des-Arg-bradykinin and des-Arg-kallidin, kinin-cleavage products are biologically inactive. Kinins are cleaved by a number of enzymes, including carboxypeptidases, ACE, and neprilysin. ACE also truncates its own reaction product, bradykinin-(1–7), further to form bradykinin-(1–5). Neprilysin, like ACE, cleaves bradykinin at the 7 to 8 position and has a broad substrate specificity (Table 11.2). The amino-terminal of bradykinin possesses two proline residues and is susceptible to cleavage by the proline-specific exopeptidase aminopeptidase P. The resultant peptide, bradykinin-(2–9), may be further cleaved by proteases that include the endothelial enzyme dipeptidyl peptidase-4, which reduces this metabolite to bradykinin-(4–9).

PLASMA AND TISSUE KALLIKREIN–KININ SYSTEM

The two independent KKSs in humans (plasma and tissue) can be distinguished by the specific subtypes of kallikreins, kininogens, and kinins involved. The circulating plasma KKS includes HMW kininogen and plasma prekallikrein, both of which are synthesized in the liver and secreted in the plasma, in which kallikrein is generated by the cell matrix–associated prekallikrein activator prolylcarboxypeptidase.[440] Of importance is that bradykinin is the main effector molecule of the plasma KKS. The tissue-specific KKS consists of locally synthesized or liver-derived kininogen (HMW and LMW), tissue kallikrein, and the effector molecules kallidin in humans and kallidin-like peptide in rodents. The half-life of kinins is 10 to 30 seconds, but in tissues with high kallikrein content, including the kidneys, local and plasma-derived LMW kininogen can be continuously cleaved to produce kallidin. Fig. 11.8 illustrates the enzymatic cascades of the plasma and tissue KKSs.

RENAL KALLIKREIN–KININ SYSTEM

The tissue KKS contributes to the physiologic functions of the kidneys with effects on RVR, natriuresis, diuresis, and other vasoactive mediators, such as renin and angiotensin, eicosanoids, catecholamines, NO, vasopressin, and ET. In the kidneys, large quantities of kininogen and kallikrein are synthesized by the tubule epithelium and are excreted in the urine. Locally formed kinin is also detectable in the urine,

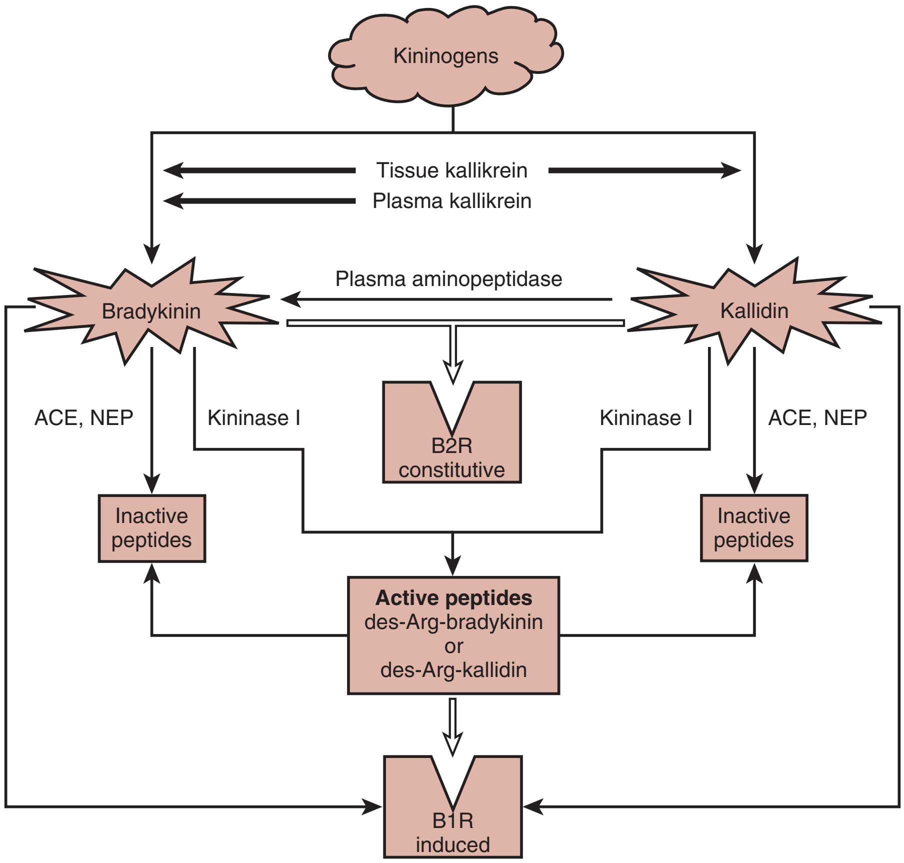

Fig. 11.8 Enzymatic cascade of the kallikrein–kinin system. *ACE,* Angiotensin-converting enzyme; *B1R,* bradykinin B1 receptor; *B2R,* bradykinin B2 receptor; *NEP,* neutral endopeptidase.

renal interstitial fluid, and renal venous blood. In the human kidneys, kallikrein is localized to the connecting tubules with close anatomic association between the kallikrein-expressing tubules and the afferent arterioles of the JGA. Results of some studies suggest that renal kallikrein mRNA is also detectable by in situ hybridization at the glomerular vascular pole. This anatomic association highlights the physiologic relationship between the KKS and the RAS and is consistent with a paracrine function for the KKS in the regulation of RBF, GFR, and renin release. In this regard, it has been suggested that, through effects on prostaglandin production, kinins may lower tubuloglomerular feedback sensitivity.

Expression of kallikrein within the kidneys is altered during development and is regulated by estrogen and progesterone, salt intake, thyroid hormone, and glucocorticoid.[441–444] The enzyme is not normally filtered at the glomerulus in the absence of glomerular injury. Kininogens are localized mostly to connecting tubule principal cells near kallikrein, which can be found in the connecting tubules of the same nephron. Once activated, renal kallikrein cleaves both HMW and LMW kininogens to release kallidin. The majority of the physiologic effects of kinins are mediated through activation of constitutively expressed B2Rs, with little or no B1R mRNA detectable in normal kidneys. In rats, administration of lipopolysaccharide, however, induces expression of B1R throughout the nephron (except the outer medullary collecting ducts), with strong expression in the efferent arteriole, medullary limb, and distal tubule.[445]

The KKS is involved in the regulation of both renal hemodynamics and tubule function. Diuretic and natriuretic effects play a pivotal role in the contribution of the renal KKS to fluid and electrolyte balance. Kinins have been reported to increase RBF and papillary blood flow and to mediate the hyperfiltration induced by a high-protein diet. Kinins also inhibit conductive sodium entry in the IMCDs,[446] and B2R-deficient mice demonstrate increased urinary concentration in response to vasopressin, which indicates that, through the B2R, endogenous kinins oppose the antidiuretic effect of vasopressin.[447] Kinins may therefore affect sodium reabsorption through direct effects on sodium transport along the nephron, through vasodilatory effects, and through changes in the osmotic gradient of the renal medulla. In addition to the effects on renal vascular tone, salt homeostasis, and water homeostasis, experiments with the B2R antagonist icatibant have yielded evidence that kinins may also have antihypertrophic and antiproliferative properties in mesangial cells, fibroblasts, and renomedullary interstitial cells. The antiproliferative effect of bradykinin in mesangial cells may be mediated through interaction of the B2R with the protein-tyrosine phosphatase SH2 domain–containing phosphatase-2.[448]

REGULATION OF TUBULE TRANSPORT BY TISSUE KALLIKREIN

Independent of its ability to generate kinin, tissue kallikrein also exerts separate effects on tubule solute transport by regulating the activity of the epithelial Na^+ channel (ENaC), the colonic H^+, K^+ ATPase, and the epithelial calcium channel TRPV5 (transient receptor potential channel vanilloid subtype 5).[449] The connecting tubules secrete a large amount of tissue kallikrein which, through its enzymatic activity, can alter the function of ion transporters expressed on the luminal surface of cells downstream of its site of secretion.[449] For instance, tissue kallikrein may participate in the proteolytic processing of ENaC increasing its activity, whereas tissue kallikrein–deficient mice have decreased ENaC activity.[450] Despite decreased ENaC activity, however, coincident upregulation of ENaC-independent electroneutral NaCl absorption ensures that tissue kallikrein is not essential for sodium homeostasis.[449,451] The cortical collecting ducts from tissue kallikrein–deficient mice also demonstrate enhanced activity of the colonic H^+, K^+ ATPase in intercalated cells, resulting in net K^+ absorption.[449,452] Finally, tissue kallikrein functions to stabilize the TRPV5 channel at the plasma membrane, promoting Ca^{2+} reabsorption, whereas tissue kallikrein knockout mice exhibit robust hypercalciuria.[453] Distal tubule defects in potassium[454] and calcium[455] handling have also been reported in humans with the loss-of-function R53H polymorphism in the tissue kallikrein gene.

THE KALLIKREIN–KININ SYSTEM IN RENAL DISEASE

HYPERTENSION

Although it has been known for many years that kinin infusion results in an acute drop in systemic blood pressure by reducing peripheral resistance, the role of the KKS in mediating primary or secondary hypertension has yet to be fully established. Decreased activity of kallikrein has been reported in the urine of hypertensive patients and hypertensive rats. An inverse relationship between urinary kallikrein excretion and blood pressure in humans may be suggestive of a role for the renal KKS in protecting against hypertension.[456] However, an alternative interpretation may be that preexisting or hypertension-induced renal disease may itself lead to a reduction in renal kallikrein excretion. In the Dahl salt-sensitive rat model of hypertension, ACE inhibitors attenuate the progression of proteinuria and hypertensive nephrosclerosis better than do ARBs.[457] That this difference may be mediated by enhanced kinin activity with ACE inhibition is supported by the observations that infusion of either kallikrein[458] or bradykinin[459] in this model attenuated glomerulosclerosis without affecting blood pressure. Studies in two-kidney, one-clip hypertension have yielded conflicting results. The incidence of two-kidney, one-clip hypertension was increased in B2R-deficient mice, in comparison with wild-type animals.[460] By contrast, with regard to the response between tissue kallikrein–deficient mice and wild-type animals, there was no difference with respect to kidney size, renin release, systemic blood pressure increase, and cardiac remodeling.[461]

Despite the uncertainty about the role of the KKS in mediating the pathogenesis of hypertension, a variety of genetic mutations of the KKS have been associated with hypertension in animal models and in humans.[462] Inactivating mutations in the kallikrein gene have been identified in spontaneously hypertensive rats,[463] and an association between mutations in the regulatory region of the kallikrein gene *KLK1* and hypertension has also been described in African Americans[464] and Chinese Han people.[465] The loss-of-function *KLK1* R53H mutation is found in 5% to 7% of the Caucasian population,[466] but this one single-nucleotide polymorphism (SNP) has not in itself been found to markedly alter blood pressure.[467] ACE polymorphisms, responsible for different plasma levels of the enzyme and, accordingly, altered kinin levels, have been identified as independent risk factors for progression of various diseases, including diabetic nephropathy, but they do not affect blood pressure. Finally, a number of SNPs in both the B2R and B1R genes have been associated with hypertension[468,469] and coronary risk in hypertensive individuals.[470]

DIABETIC NEPHROPATHY

Observations in both experimental animal models and in humans indicate a role for altered KKS activity in the pathogenesis of diabetic nephropathy, although results in experimental studies have been conflicting. The KKS is markedly altered in rats with streptozotocin-induced diabetes and changes are correlated with those in renal plasma flow and GFR.[471] Renal and urinary levels of active kallikrein are increased in rats with moderate hyperglycemia in association with reduced RVR, increased GFR, and increased renal plasma flow and treatment of diabetic rats with the kallikrein inhibitor aprotinin or with a B2R antagonist reduced RBF and GFR.[471] By contrast, in non–insulin-treated streptozotocin-treated rats with severe hyperglycemia and hypofiltration, kallikrein excretion and expression were reduced.[471,472] In addition to its hemodynamic effects, the KKS may also play a renoprotective role in diabetic nephropathy through its antiinflammatory and antiproliferative properties.[424]

Results of receptor antagonist studies initially suggested that the KKS had a limited role in preserving renal structure and function in diabetic nephropathy: Treatment of diabetic rats with icatibant had no effect on glomerular structure or on albuminuria, nor did it alter the attenuating effect of ACE inhibition on either of these parameters.[473] In contrast to this finding, however, the results of more contemporary work suggest that the beneficial effects of ACE inhibitors in experimental diabetic nephropathy may be attenuated by coadministration of a B2R antagonist.[474–476] In Akita diabetic mice lacking the B2R, there was a marked increase in mesangial sclerosis and a worsening of albuminuria,[477] in association with an increase in oxidative stress and mitochondrial damage[478]; however, another study reported contrary results in that B2R-knockout mice were relatively protected from the renal injury caused by streptozotocin-induced diabetes.[479] Upregulation of B1R occurs in response to B2R knockout and could plausibly contribute to renal pathology or alternatively confer a renoprotective benefit. In support of the latter thesis, Akita-diabetic mice deficient in both B2R and B1R exhibited augmented renal injury in comparison to those lacking B2R alone.[480]

In further support of a renoprotective effect of the KKS in diabetic nephropathy, one study showed that induction of diabetes by streptozotocin in mice caused a twofold increase in mRNA for kininogen, tissue kallikrein, kinins, and kinin receptors, with a doubling in albumin excretion in kallikrein-knockout mice in comparison with wild-type animals.[481] In another study, gene delivery of human tissue kallikrein with an adeno-associated virus vector attenuated renal injury in diabetes and decreased urinary albumin excretion.[482] When exogenous pancreatic kallikrein was administered to diabetic mice, it caused a reduction in albuminuria, renal fibrosis, inflammation, and oxidative stress.[483] Conversely, however, kallistatin, which decreases kallikrein activity, also attenuated renal injury in diabetic mice when it was overexpressed using ultrasound microbubble-mediated gene transfer.[484]

Urinary kallikrein excretion in patients with type 1 diabetes demonstrates a similar association with GFR as observed in rats with streptozotocin-induced diabetes.[485] Active kallikrein excretion is increased in hyperfiltering individuals in comparison with both patients with type 1 diabetes who have a normal GFR and normal controls, and it is correlated with both GFR and distal tubule sodium reabsorption.[485] Results of genetic association studies in patients with diabetes have, however, been conflicting: One study demonstrated an association between B2R polymorphisms and albuminuria in 49 patients with type 1 diabetes and 112 patients with type 2 diabetes,[486] whereas another revealed no association between either B1R or B2R polymorphisms and incipient or overt nephropathy in 285 patients with type 2 diabetes.[487] Plasma levels of HMW kininogen fragments were observed to be elevated among individuals with type 1 diabetes and progressive renal decline.[488]

ISCHEMIC RENAL INJURY

In models of ischemia-reperfusion injury, ACE inhibitors appear to be superior to ARBs in protecting against tubule necrosis, loss of endothelial function, and excretory dysfunction.[489] This superiority may be attributed to enhanced kinin activity with ACE inhibition, inasmuch as the effect is negated by B2R antagonists and inhibitors of NO synthase.[490,491] Bradykinin suppresses the opening of mitochondrial pores,[492] and NO suppresses oxidative metabolism; both observations indicate that the KKS may exert its protective effects in ischemia-reperfusion injury through attenuation of oxidative damage. In mice with genetic deficiencies in either the B2R alone or both B1R and B2R, ischemic damage was enhanced in comparison with wild-type mice; injury was most severe in mice that lacked both receptors.[436] By contrast, tissue kallikrein infusion aggravated renal ischemia-reperfusion injury in rats,[493] whereas expression of the human kallistatin gene with an adenoviral vector protected mice from renal ischemia-reperfusion injury.[494] Thus although physiologic kinin levels may be protective in this setting, higher levels may be detrimental, possibly through pathologic reperfusion.[422]

CHRONIC KIDNEY DISEASE

In the remnant kidney model of progressive renal disease, adenovirus-mediated or adeno-associated virus–mediated gene delivery of kallikrein attenuated the decline in renal function.[495] In the model of unilateral ureteric obstruction, both genetic ablation of the B2R and pharmacologic blockade of the B2R increased tubulointerstitial fibrosis.[496] By contrast, expression of the B1R is increased after unilateral ureteric obstruction,[497] and treatment with a nonpeptide B1R antagonist reduced macrophage infiltration and fibrosis.[497] In the same model, B1R-deficient mice similarly showed less upregulation of inflammatory cytokines, reduced albumin excretion, and diminished fibrosis in comparison with wild-type mice.[498] In an Adriamycin-induced mouse model of FSGS, B1R antagonist therapy attenuated and B1R agonist therapy aggravated renal dysfunction.[499] Together, these observations suggest that although the B2R is renoprotective, under some circumstances (and in contrast to the observations made in Akita-diabetic B1R/B2R knockout mice) compensatory B1R upregulation may contribute to the pathogenesis of renal fibrosis. In humans, polymorphisms in both the B1R gene[500,501] and the B2R gene[500,502] have been associated with the development of ESRD.

LUPUS NEPHRITIS/ANTI–GLOMERULAR BASEMENT MEMBRANE DISEASE

Evidence has linked the KKS to the pathogenesis of the immune-mediated nephritides, SLE, Goodpasture syndrome (antiglomerular basement membrane [GBM] disease), and spontaneous lupus nephritis. Mice strains differ in their susceptibility to anti-GBM antibody–induced nephritis. Comparison of disease-sensitive and control strains, by microarray analysis of renal cortical tissue, revealed that 360 gene transcripts were differentially expressed.[503] Of the underexpressed genes, one-fifth belonged to the kallikrein gene family.[503] Furthermore, in disease-sensitive mice, B2R antagonism augmented proteinuria after anti-GBM challenge, whereas bradykinin administration attenuated disease.[503] In the same study, SNPs in the *KLK1* and *KLK3* promoters were also described in patients with SLE and lupus nephritis.[503] Extending their work further, the same investigators showed that adenoviral delivery of the *KLK1* gene attenuated renal injury in congenic mice possessing a lupus-susceptibility interval on chromosome 7.[504]

ANTINEUTROPHIL CYTOPLASMIC ANTIBODY–ASSOCIATED VASCULITIS

Granulomatosis with polyangiitis (GPA) may be associated with a necrotizing glomerulonephritis. The major antigenic target in GPA is neutrophil-derived proteinase 3 (PR3). Incubation of PR3 with HMW kininogen resulted in the generation of a novel tridecapeptide kinin, termed "PR3-kinin."[505] PR3-kinin binds to B1R directly and can also activate B2R after further processing to form bradykinin.[505] These observations suggest that, in GPA, PR3 may activate the kinin pathway in a kallikrein-independent manner. B1R upregulation has been observed in biopsies from patients with Henoch–Schönlein purpura nephropathy or with antineutrophil cytoplasmic antibody–associated vasculitis.[506] Similarly, B1R upregulation was also observed in a murine serum–induced glomerulonephritis model, whereas treatment with a B1R antagonist attenuated renal decline.[506]

UROTENSIN II

Urotensin II (U-II) is a potent vasoactive cyclic undecapeptide originally isolated from the caudal neurosecretory organ of teleost fish. The system is now known to be present in humans. The two principal regulatory peptides derived from this organ are urotensin I (U-I), which is homologous to mammalian corticotropin-releasing factor, and U-II, which bears sequence similarity to somatostatin[507] and has notable hemodynamic, gastrointestinal, reproductive, osmoregulatory, and metabolic functions in fish. Homologs of U-II have been identified in many species, including humans.

SYNTHESIS, STRUCTURE, AND SECRETION OF UROTENSIN II

Human U-II is derived from two prepropeptide alternate splice variants of 124 and 139 amino acids, differing only in the N-terminal sequence.[508,509] The C terminus is cleaved by prohormone convertases to yield the mature 11–amino acid U-II peptide. U-II contains a cyclic Cys-Phe-Trp-Lys-Tyr-Cys hexapeptide sequence that is conserved across species and is essential for its biologic activity[510] (Fig. 11.9). The N-terminal region of the precursor is highly variable across species. Prepro–U-II mRNA has been described in a range of cell types, including vascular smooth muscle cells, endothelial cells, neuronal cells, and cardiac fibroblasts. Multiple monobasic and polybasic amino acid sequences have been identified as posttranslational cleavage sites of the prohormone. However, a specific U-II–converting enzyme has not yet been described. With respect to its tissue distribution, immunohistochemical staining has identified U-II protein in the blood vessels of various organs and also within the tubule epithelial

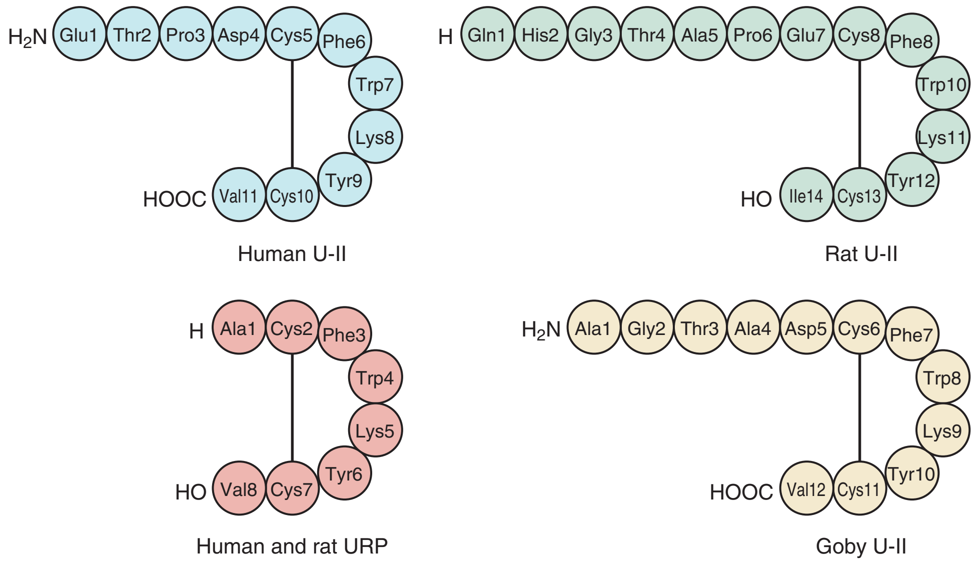

Fig. 11.9 Molecular structure of human, rat, and goby urotensin II (U-II). *URP,* Urotensin-related peptide. (Modified from Ashton N. Renal and vascular actions of urotensin II. *Kidney Int.* 2006;70:624–629.)

cells of the kidneys.[507,511,512] A significant arteriovenous gradient exists across the heart, liver, and kidneys, which indicates that these organs are important sites of U-II production.[513]

In 1999, Ames and colleagues[509] identified U-II as the ligand for the previously orphan rat receptor GPR14/SENR. The U-II receptor (commonly referred to as the UT receptor) is a seven-transmembrane, GPCR, encoded on chromosome 17q25.3 in humans,[514] that bears structural similarity to both somatostatin receptor subtype 4 and the opioid receptors. Ligand binding of the receptor results in G protein–mediated activation of PKC, calmodulin, and phospholipase C; evidence also links MAPKs ERK1/2, the Rho kinase pathway, and peroxisome proliferator-activated receptor α in the intracellular signaling cascade.[515–518]

The relationship between U-II and the UT receptor is not exclusive; the receptor also binds alternative U-II fragments such as U-II(4-11) and U-II(5-11), as well as urotensin-related peptide (URP).[519,520] URP was originally isolated from rat brain and binds with high affinity to the UT receptor.[520] Although this 8–amino acid peptide retains the cyclic hexapeptide sequence, it is derived from a different precursor to U-II and may have different physiologic properties.[521]

PHYSIOLOGIC ROLE OF UROTENSIN II

U-II is the most potent vasoconstrictor known, being 16 times more potent than ET-1 in the isolated rat thoracic aorta.[509] However, its vasoconstrictive properties are not universal, varying between species and between vascular beds. For example, U-II has little or no effect on venous tone, and it does not cause constriction of rat abdominal aorta, femoral arteries, or renal arteries.[522] It also lacks systemic pressor activity when administered intravenously to anesthetized rats.[509,523] In cynomolgus monkeys, bolus intravenous injection of U-II induced myocardial depression, circulatory collapse, and death.[509] In contrast to the vasoconstrictive properties of vascular smooth muscle UT receptor, endothelial UT receptor may mediate vasodilation in pulmonary and mesenteric vessels.[524] The response to U-II may be dependent on the caliber of the artery; a small-vessel response is more endothelium mediated, and a large-vessel response is more dependent on vascular smooth muscle.[507] These disparities are among many examples of how the role of U-II may be influenced by a number of factors, including animal model, vascular bed, method of exogenous U-II administration, and the presence of comorbid conditions.

UROTENSIN II IN THE KIDNEY

The kidneys are major sites of U-II production; this is indicated by both the arteriovenous gradient of plasma U-II across the kidneys and the observation that urinary U-II clearance exceeds urinary creatinine clearance.[507,513] In fact, in humans, urinary concentrations of U-II are approximately three orders of magnitude higher than plasma concentrations.[525] U-II is present in a number of kidney cell types, including the smooth muscle cells and endothelium of arteries, proximal convoluted tubules, and particularly the distal tubules and collecting ducts.[512] UT receptor mRNA is also present in the kidneys, especially within the renal medulla,[525–527] which suggests that the peptide may have autocrine or paracrine functions at this site. In addition, URP mRNA has been described in both rat and human kidneys.[520,526,527] Studies of the role of U-II in normal renal physiology have yielded conflicting findings. In one report, continuous infusion of U-II into the renal artery of anesthetized rats caused NO-dependent increases in GFR, urinary water excretion, and urinary sodium excretion.[528] By contrast, another study showed that bolus injection of picomolar concentrations of U-II produced a dose-dependent decrease in GFR and a reduction in urine flow and urinary sodium excretion.[527] Furthermore, a third group of researchers reported that intravenous bolus injection of U-II in nanomolar amounts induced only a minor reduction in GFR and had no effect on sodium excretion.[529] These researchers also investigated the effect of U-II administration in the context of experimental congestive heart failure, in which the peptide induced an almost 30% increase in GFR.[529]

OBSERVATIONAL STUDIES OF UROTENSIN II IN RENAL DISEASE

Variations in the concentration of U-II in the plasma and urine have been found in a number of diseases with, again, sometimes conflicting results. U-II levels in plasma may be increased in hypertensive individuals in comparison with normotensive controls and are correlated with systolic blood pressure.[530] In one study, U-II concentrations in plasma were increased twofold in patients with renal disease not on hemodialysis and threefold in patients on hemodialysis.[531] In a separate study, the same investigators observed U-II levels in both plasma and urine to be higher in patients with type 2 diabetes and renal disease than in such patients with normal renal function.[532] Higher U-II levels in urine have been described in patients with essential hypertension, in patients with glomerular disease and hypertension, and in patients with renal tubule disorders, but not in normotensive patients with glomerular disease.[525] Increased expression of both U-II and the UT receptor have also been demonstrated in biopsy samples of patients with diabetic nephropathy[533]; increased U-II levels have also been described in glomerulonephritis[534] and in minimal change disease.[535] By contrast, a more recent study described a reduction in U-II levels with CKD. Here, investigators reported that plasma U-II concentrations were highest in healthy individuals, lower in individuals with ESRD, and lowest in subjects with non-ESRD CKD, while hypothesizing that the discordance with earlier work may reflect the different populations studied or the assays used.[536]

INTERVENTIONAL STUDIES OF UROTENSIN II IN THE KIDNEY

Both peptide and nonpeptide UT receptor antagonists have been studied. Urantide is a derivative of human U-II.[537] Continuous infusion of urantide into rats induces an increase in GFR and natriuresis,[527] although it is not clear whether the natriuresis is a consequence of altered renal vascular tone or a direct effect of U-II on the tubule epithelium. Whereas urantide is a potent antagonist of the rat UT receptor,[537] it has been found to have agonist properties in cells expressing the human UT receptor.[538] An alternative U-II peptide antagonist, UFP-803, also has partial agonist properties in human UT receptor–expressing cells,[539] which complicates the interpretation of a peptide-based approach to U-II inhibition. Two compounds in the nonpeptide group of U-II antagonists have been studied: palosuran (ACT-058362) and SB-611812. Intravenous administration of palosuran protected against renal ischemia in a rat model.[540] The same compound was also studied in rats with streptozotocin-induced diabetes, in which it was found to significantly reduce the severity of albuminuria.[541] In a study of 19 individuals with type 2 diabetes and macroalbuminuria, palosuran attenuated urine albumin excretion after 2 weeks.[542] However, in a subsequent 4-week study of 54 individuals with type 2 diabetes, hypertension, and nephropathy, palosuran had no effect on albuminuria, blood pressure, GFR, or renal plasma flow,[543] effectively halting the development of this drug for this indication. SB-611812 decreased the carotid intima-to-media ratio in a rat model of balloon angioplasty–induced stenosis.[544] The same compound attenuated myocardial remodeling and was associated with a reduced rate of mortality in a rat model of ischemic cardiomyopathy.[545,546] At present, there are no reports of the effect of SB-611812 in renal disease.

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BOARD REVIEW QUESTIONS

1. The predominant site for angiotensinogen synthesis is?
   a. The juxtaglomerular apparatus
   b. Vascular endothelium
   c. Pericentral zone hepatocytes
   d. Peritubular interstitial fibroblasts

   Answer: c

   Rationale: The RAS is a multi-organ endocrine system wherein specific cell types have key roles. Understanding which cell in which organ does what is an integral part of knowing how the system works.

2. Acute angiotensin type 1 receptor stimulation causes all but one of the following?
   a. Elevated intraglomerular pressure (P_{GC})
   b. Reduction in the ultrafiltration coefficient (K_f)
   c. Proteinuria
   d. Increased afferent and efferent arteriolar tone

   Answer: c

   Rationale: The effects of angiotensin II and the activation of its type 1 receptor on glomerular hemodynamics is pivotal in understanding the role of the RAS in physiology, pathophysiology and therapeutics.

3. Concerning angiotensin-converting enzyme 2, which of the following are true?
   a. Shares >85% homology with angiotensin-converting enzyme in terms of its amino acid sequence
   b. Cleaves angiotensin II to angiotensin III
   c. Is the receptor for the severe adult respiratory syndrome (SARS) coronavirus
   d. Within the glomerulus is expressed predominantly in mesangial cells

   Answer: c

   Rationale: Angiotensin converting enzyme 2 is a relatively new but important addition to the RAS. Knowledge of its origin and actions are necessary to fully appreciate the complexites of the system.

4. B-type natriuretic peptide binds to which of the following receptors?
   a. Natriuretic peptide receptor-A
   b. Natriuretic peptide receptor-B
   c. Natriuretic peptide receptor-B and natriuretic peptide receptor-C
   d. Natriuretic peptide receptor-A and natriuretic peptide receptor-C

   Answer: d

   Rationale: The nomenclature for the natriuretic peptide receptors can be confusing. NPR-A binds ANP and BNP. NPR-B binds CNP and NPR-C is a clearance receptor for all three peptides.

5. Which of the following is not a known substrate of neprilysin?
   a. Vascular endothelial growth factor-A
   b. Amyloid β-peptide
   c. C-type natriuretic peptide
   d. Vasopressin

   Answer: a

   Rationale: Neprilysin is a zinc metalloproteinase that catalyzes the degradation of the natriuretic peptides and that has activity against over 40 other substrates (Table 11.2). Vascular endothelial growth factor-A is not amongst the reported substrates of neprilysin.